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Open API Map

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Executive Summary

In this increasingly digitized world, IT technology is connecting more and more people. This technology brings the enterprises into a dynamic, distributed environment and delivers business value on multiple channels including desktops and mobile devices. Since the complexity and scale is unprecedented, more openness and agility is required during innovation and competition in today's economy. The industrial trend has shown clearly that enterprises must provide more visibility to their APIs (Application Programming Interfaces) to facilitate open integration.

The TM Forum knows the importance of APIs and a group of open digital API definitions has been made. To date, those definitions have been based on the contributions of discrete API initiatives.

The API Map is intended to show the complete landscape of all the necessary APIs in a telecommunications ecosystem. It can guide the Service Providers or other large Enterprises to construct the ecosystem to attract DSP partners and integrating with them or as a basis of customization for individual partnerships.

The API Map will also uncover the future-oriented planning for potential new business APIs. In the years hereafter, those API definitions will be explored and the efforts to build them are encouraged fully.

The use of a standard API Map will facilitate the understanding of the interoperability between Service Providers and their partners and provide additional benefits for the enterprises such as:

- With a unified API Map, an enterprise can experience:
 - o Lower software integration costs, as application vendors converge on standardized exposure of capabilities
 - o Increased portability of applications
 - o Improved interoperability within and between business functions
 - o Easier upgrade and exchange of system components.
- By exposing IT capabilities as APIs, we will find new partnership models to enhance our services.

1. Introduction

This document provides an overview of the TM Forum API Map and its inheritance from the TM Forum API done to-date.

Its purpose is to present the value of the TM Forum API Map which shows the complete picture of business related API for the first time.

It is structured as follows:

- An overview of the Open Digital API Map:
 - o The business value it can provide
 - o The principles of the API Map
- A hierarchical taxonomy of the API Map
- A description of all the current TM Forum APIs in the API Map
- A summary of planned and future APIs which are in development now or need to be developed in future within the TM Forum.

1.1. The audience of the API Map

This document is intended for any party wishing to play a role in the open digital economy – Commercial, Operations and Technical management in Communications and Digital Service Providers, Content Providers and Brokers, Software Suppliers and Systems Integrators – who want to use/bundle simple content/communication services traditionally only available to Telco/Cable providers & partners.

It should have optimal benefit for players in industries outside ICT (example Automotive, Health Care ...), or for small technology startups that cannot afford the high investment in bespoke enterprise integration.

The specific focus is on business people wishing to know what they can achieve by using TM Forum APIs. In addition, IT Architects and developers will benefit from an overview of available APIs.

2. Open Digital API Map Overview

2.1. Business value of API Map

The API Map is helpful in a variety of possibilities to create distinct values. By building API following the map, the digital ecosystem can be achieved with fewer obstacles and it will result in systems integration and data sharing between partners through API interactions:

- For vertical industries
 - o Reduce the difficulties to enter and combine communication capabilities with the creative products, spanning national borders and specific brands
- For start-ups and SME
 - o Availability of a simple and modern interface technology requiring minimum domain expertise and allowing quick setup and deployment.
- For Telco & Information Providers
 - o Reduce administration of multiple external disparate interfaces.
 - o Faster capitalization of assets (increased usage through micro and cross industry exposure).

2.2. Principles of API Map

There are a number of industry-standard principles that guide the construction of good APIs, such as:

- **Abstraction:** Reflect the “business” service being provided, hiding the implementation.
- **Loose-coupling:** Do not expose any technical dependencies/intimate knowledge of the underlying implementation.
- **Reusability:** APIs are defined with re-use in mind (even for unknown future use-cases).
- **Discoverability:** Service contracts can be effectively communicated and interpreted.
- **Developer-friendly:** APIs can be consumed with minimal effort/cost.

Especially for the API in this Map, the division and position of the APIs also follow the distinct organization principles, as follows:

- Each API belongs to one or several Level 0 domains rooted in TMF Information Framework (SID) model.
- The domain of the API is determined by the relevant Information Framework ABE which is the base of the main resource of that API.
 - o If there are multiple Information Framework ABE entities which are depended by one API resource, the main Information Framework ABE is chosen according to the degree of correlation between Information Framework ABE and API resource.
 - o If there are multiple Information Framework ABE entities which are depended by one API resource, and the Information Framework ABE entities have same degree of importance for this API resource, such API should be placed in all relevant domains.

- The relationship between API and Information Framework will still be verified via business process (As shown in Chapter 4, this will be done in the future).
- The existing published TMF API is expected to keep as it is currently. Any change to the existing APIs should be discussed and agreed.
- The API should contain the tightly-related resources for the same purpose. The loose-coupled resources need to be divided into different API.
- The purpose of API and exposed resources in the API should not be duplicated.

2.3. Consistency with API Catalog

The API Catalog is regarded as the unique representation of TM Forum APIs for the members to trace all the APIs which has been confirmed by API Governance Board.

This Map keeps the consistent information with the API Catalog, but it contains the levels and detailed descriptions for those API. Additionally, the map provides the landscape for the API evolution. It also gives the examples for the usage of APIs to help the members to understand and apply them.

3. Hierarchy of the API & Resource Map

The TMF APIs can be related to all other TMF frameworks, in that they:

- Are typically entity-centric, manipulating a primary resource that can be found in TMF Information Framework, an Information view of the API's
- Play a part in (potentially a number of) business processes from TMF-BPF (eTOM), a Business Process view of the API's
- Can be exposed by a normalized endpoint application or application-domain described by TMF-TAM, an Application view of the API's

You can therefore consider the API Map as a central catalog with a number of framework-centric views, expressed as mappings into that catalog.

Multiple views of the API Map are required to address different concerns and to represent different viewpoints.

The Information view of the API Map shows the API and API resources mapped to the Information Framework data domains and ABE's. This is useful to understand what data *is managed* through which API's.

The Business Process view of the API Map shows the API's in business context. This is useful to understand how API's *are used* to support the business processes of an organisation.

The 'Information view' of the API Map is organized in multiple levels:

- Level 0: the standard Framework business domains
- Level 1: The main TM Forum APIs, typically the primary resource, and aligned to TMF Information Framework ABEs.
- Level 2: the component resources exposed within the API.

3.1. Level 0 Map

The general business capabilities are divided according to SID domain:

Marketing/Sales	The Market/Sales domain supports the sales and marketing activities needed to gain business from customers and potential customers.
Product	The Product domain is concerned with the lifecycle of products offered to and procured by customers.
Customer	The Customer domain represents individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data.
Service	The Service Domain is concerned with the definition, development, and operational aspects of Services used to realize offerings to the market.
Resource	The Resource domain is concerned with the definition, development, and operational aspects of the applications, computing, and networks which represent the infrastructure of an enterprise.
Engaged Party	The Engaged Party domain encompasses planning of strategies for Engaged Parties, handling of all types of contact with Engaged Parties, the management of the relationship, and the administration of Engaged Parties data. I

Enterprise	The Enterprise domain provides support and sets policy for the overall business, enterprise or Service Provider. It also includes activities that are common to all enterprises across all industries such as accounting and human resource management.
Common	The common domain is of different nature from the above defined ones as the processes, information data and applications described there do not necessarily have relationships.

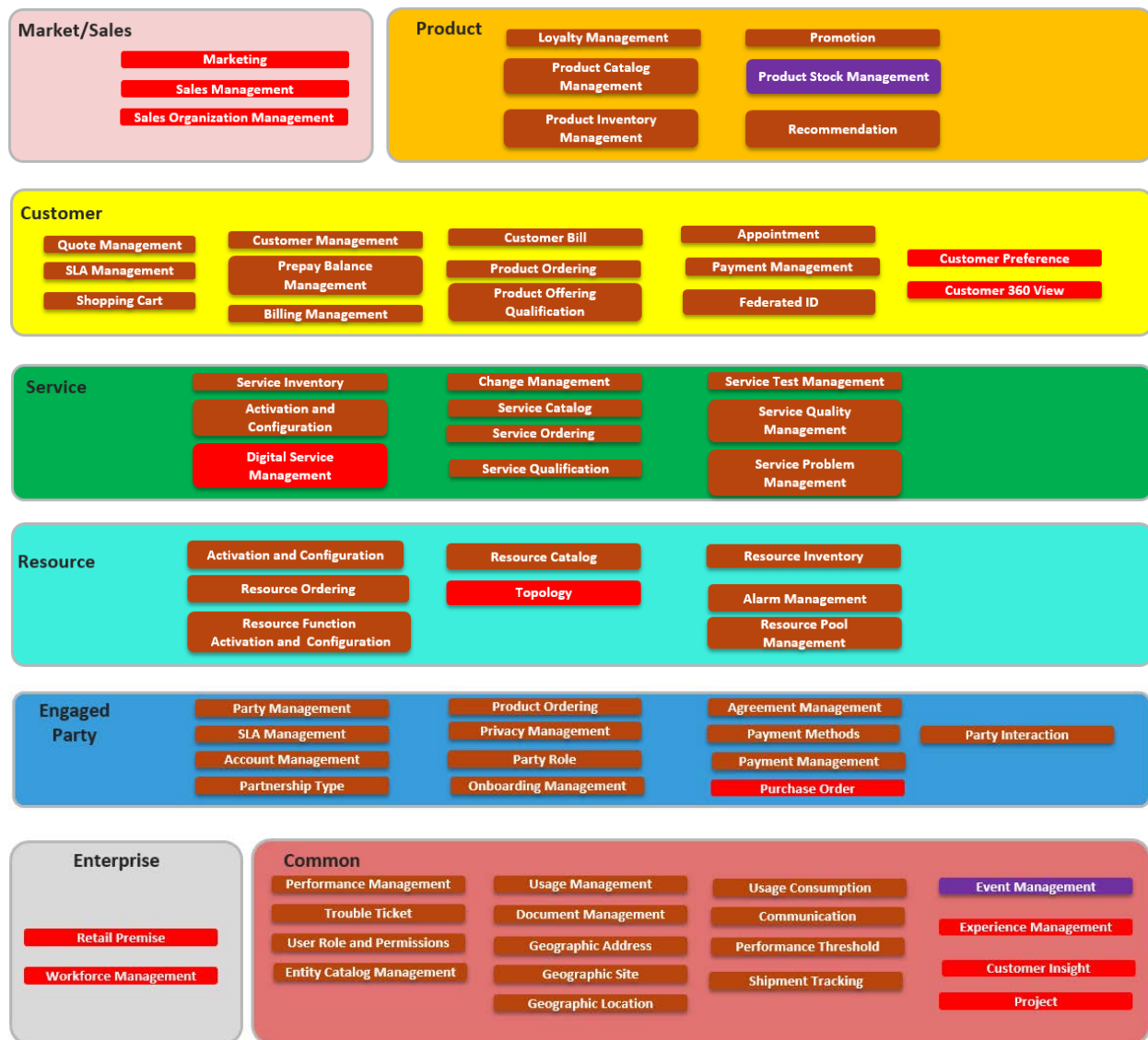


Figure 1 – Level 0 Map

3.2. Level 1 Map

Level 1 is the main TM Forum APIs and aligned to TMF-Information Framework ABEs. Each level-0 part in the API Map shall contain the APIs which are based on the enterprise business operation requirements.

An API can contain a group of API resources which are aggregated to accomplish the same business purpose. API resource is mapped in Level 2 of the API Map.



The legend of the API status is as the following:

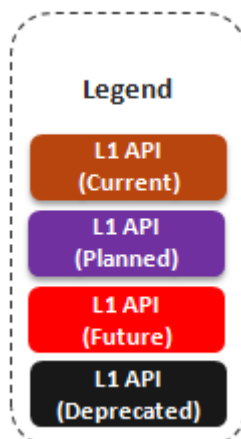


Figure 2 – Level 1 Map

Meaning of the status:

- “Current” : The API has been published with approved specification.
- “Planned”: The API has the draft specification which is still under reviewing of the stakeholders.
- “Future”: The API is recognized as useful, but no specification has been accepted by the API Governance board to start the reviewing.
- “Deprecated”: The API has been once published. However, it is replaced by other new API or become not required.

Note:

1) Product Ordering API, SLA API, Activation and Configuration, , Payment Management API appear in multiple domains due to the Information Framework mapping relationship.

2) “Address” API will be renamed as “Geographic Address”

3.2.1. Level 1 Map – Current & Deprecated API's

The following is a view of all existing TMF APIs:

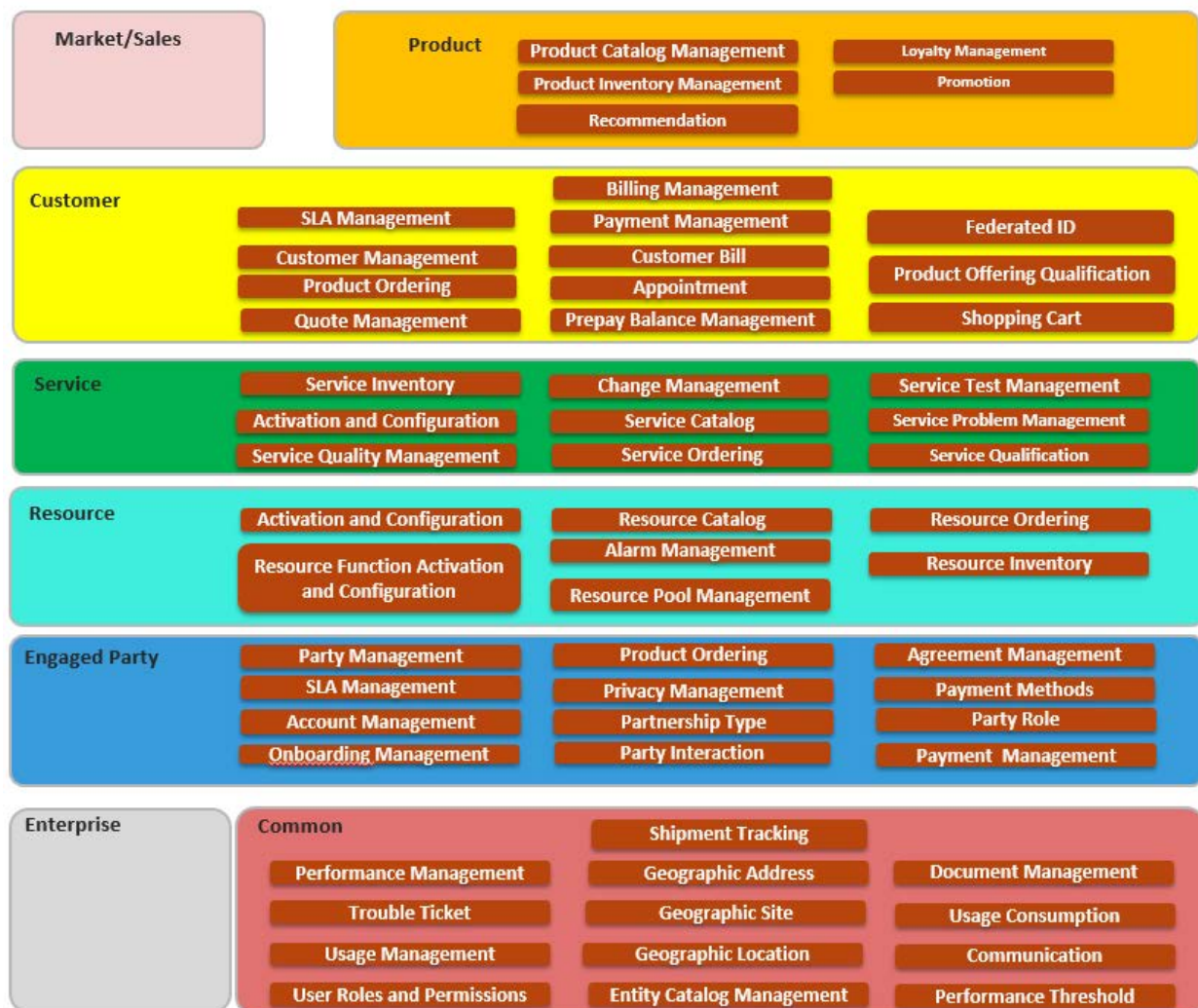


Figure 3 - API Map Level 1 (Current/Existing & Deprecated)

Note: Product Ordering API, SLA API, Activation and Configuration API appear in multiple domains due to the SID mapping relationship.

To date, the TM Forum Open Digital API has delivered or started defining the following APIs:

- Billing Management API

The Billing Management API provides standardized mechanisms for billing account, bill item and settlement note advice management either in B2B or B2B2C contexts. It allows creation, update and retrieval of a billing account either in a B2B2C relationship context (creation of mass market customer billing account within a “Billing on Behalf of” process for example) or in a B2B context (creation of a billing account for a partner or B2B customer). It allows also creation and query of bill items allowing partners or B2B customer to check their invoice. In addition, it allows notification of settlement note advice to partners who can, then, query it. There are plans to improve the billing management API. For 16.5 everything is going to stay as it is in the billing domain.

- Promotion API

Promotion is the online incentive which provides discount, gift or other substantial stimulation to encourage more consumption according to the purchased offering.

- Shopping Cart API

This API provides a shopping cart which is used for the temporarily selection and reservation of offerings in e-commerce and retail purchase.

- Product Ordering API

The Product Ordering API provides a standardized mechanism for placing a product order with all of the necessary order parameters. The API consists of a simple set of operations that interact with CRM/Order negotiation systems in a consistent manner. A product order is created based on a product offering that is defined in a catalog. The product offering identifies the product or set of products that are available to a customer, and includes characteristics such as pricing, product options and market.

The product order references the product offering and identifies any specific requests made by the customer.

- Product Catalog Management API

The Product Catalog Management API allows the management of the entire lifecycle of the catalog elements, the consultation of catalog elements during several processes such as ordering process, campaign management and sales management.

- Loyalty Management API

A loyalty program product specification is a detailed description of a loyalty program made available externally in the form of a Loyalty Product to Loyalty Program Members. A loyalty program product specification defines one or more Loyalty Rules that have to be checked in order to identify the actions to apply

- Trouble Ticket API

The Trouble Ticket API provides a standardized client interface to Trouble Ticket Management Systems for creating, tracking and managing trouble tickets among partners as a result of an issue or problem identified by a customer or another system. Examples of Trouble Ticket API clients include CRM applications, network management or fault management systems, or other trouble ticket management systems (e.g. B2B).

The API supports the ability to send requests to create a new trouble ticket specifying the nature and severity of the trouble as well as all necessary related information. The API also includes mechanisms to search for and update existing trouble tickets. Notifications are defined to provide information when a ticket has been updated, including status changes. A basic set of states of a trouble ticket has been specified to handle ticket lifecycle management.

- SLA Management API

The SLA API provides a standardized interface for SLA life cycle Management (SLA Negotiation, SLA configuration SLA Activation/enforcement, SLA Operations, SLA violation / consequence handling, SLA reporting) between a Customer and a Service Provider which provides offers (product with attached SLA in its catalog) the customer can discover, browse, trigger and order.

It also will be also useful in a multi-partner environment where exchanging SLA is needed in order to allow rapid and efficient SLA life cycle management across partners' environment. From SLA perspective, duties and rights are assigned to each actor & associated roles mainly in the case where a service is composed of various components brought by different partners within federation or / and syndication models.

- Performance Management API

The Performance Management API provides standardized mechanism for performance management such as creation, partial or full update and retrieval of the resources involved in performance management (Measurement Production Job, Measurement Collection Job, and Ad hoc Collection). It allows also notification of events related to performance.

- Customer Management API

The Customer Management API provides standardized mechanism for customer and customer account management, such as creation, update, retrieval, deletion and notification of events.

Customer can be a person, an organization or another service provider who buys products from an enterprise. Customer management API allows management of identification and financial information about him.

- Party Management API

The Party Management API provides standardized mechanism for party management such as creation, update, retrieval, deletion and notification of events.

Party can be an individual or an organization that has any kind of relation with the enterprise.

Party is created to record individual or organization information before the assignment of any role.

For example, within the context of a split billing mechanism, Party management API allows creation of the individual or organization that will play the role of 3rd party payer for a given offer and, then, allows retrieval or update of their information.

- Usage Management API

The Usage management API provides standardized mechanism for usage management such as creation, update, retrieval, import and export of a collection of usages.

This API will be extended to include more services relevant to usage charging.

Usage API manages both rated and non-rated usage.

For example, Usage API allows a service provider:

- To retrieve usage generated by a partner service platform in order to rate it,
- To provide rated usage to a partner for consumption follow up purposes.

- Product Inventory Management API

The Product Inventory API provides standardized mechanism for product inventory management such as creation, partial or full update and retrieval of the representation of a product in the inventory. It also allows the notification of events related to product lifecycle.

For example, product inventory API can be used to retrieve products owned by a customer or update the status of an installed product.

- Activation and Configuration API

The Activation and Configuration API allows the user to retrieve, create, update, delete services and retrieve the monitor resource used to monitor the execution of asynchronous requests on specific resource.

- Privacy Management API

The Privacy Management API provides the management for the customer privacy information and preference.

- Onboarding Management API

It provides standardized mechanisms for managing an onboarding process. The intention for onboarding process in the Digital Ecosystem is to have a lightweight approach similar to an end-user signing-on to terms and conditions for downloadable applications. The interface will provide onboarding that identifies that a new API would be needed to automate the onboarding process. The onboarding of the “Party”, the role can be Partner, Supplier, Developer, etc. The onboarding of the “Services” could be product offerings.

- Agreement Management API

This API provides standardized mechanism for managing agreements, especially in the context on partnerships between partners.

- Service Qualification API

Service Qualification API is one of the Pre-Ordering Management APIs. Service Qualification API goal is to provide service availability at Customer location.

- Appointment API

The Appointment API is one of the Pre-Ordering Management APIs. The appointment API provides a standardized mechanism to book an appointment with all the necessary appointment characteristics. First, the API consists in searching free slots based on parameters, as for example a party. Then, the appointment is created. The appointment has characteristics such as nature of appointment, place of appointment.

- Quote API

The Quote API is one of the Pre-Ordering Management APIs. The customer Quote API provides a standardized mechanism for placing a customer quote with all of the necessary quote parameters.

- Prepay Balance Management API

Prepay Balance Management API manages the balance, recharge (top-up) and transfer resources.

- Service Quality Management API

This API provides Quality Management for the certain service. It enables easy integration of Service Quality Management applications and client applications within the Digital Eco-system, where Service Quality Management application may reside in one enterprise and client applications may be in multiple other Enterprises.

- Service Test Management API

This API provides test procedure for the certain service. The Service Test API provides a standardized mechanism for placing a service test with all of the necessary test parameters.

- Change Management API

This API provides the control mechanism for the change of the service. Change Management process is to respond to the customer's changing business requirements while maximizing value and reducing incidents, disruption and network. The Change Management API provides the standard integration capabilities between external applications and Change Management Application.

- Service Catalog API

This API allows the management of the entire lifecycle of the catalog elements, the consultation of catalog elements during service processes.

- Service Inventory

The Service Inventory API can be used to query the service instances for a customer via Self Service Portal or the Call Centre operator can query the service instances on behalf of the customer while a customer may have a complaint or a query.

- Service Ordering API

The Service Ordering API provides a standardized mechanism for placing a service order with all of the necessary order parameters. It allows users to create, update & retrieve Service Orders and manages related notifications.

- Resource Ordering API

This API provides the order implementation on the resource to change or allocate the resource to meet the request of product order and service order.

- Resource Function Activation& Configuration API

This API is to be used for provisioning of components in support of cloud services. The components may be virtualized or physical.

- Service Problem Management API

This SPM API is used for the service providers (Defined as the Middle B) to manage the service problems in their service area. Service problem is generated based on the information declared by Middle B or the event information notified from infrastructure providers (Defined as the First B) who provide the infrastructure of cloud or network.

- Resource Catalog API

The Resource Catalog Management API REST specification allows the management of the entire lifecycle of the Resource Catalog elements, the consultation of resource catalog elements during several processes such as ordering process, campaign management, and sales management.

- Document Management API

It provides the operations to synchronize documents and document versions across systems. It also provides operations for uploading documents by Users as well as for viewing of documents online.

- Payment Methods API

This API supports the frequently-used payment methods for the customer to choose and pay the usage, including voucher card, coupon and money transfer.

- Account Management API

The Account API provides standardized mechanism for the management of billing and settlement accounts, as well as for financial accounting (account receivable) either in B2B or B2B2C contexts.

- Party Role API

Party role is the basis data for the customer and partner. It is the business incarnation of the party.

- Partnership Type API

The Partnership Type API provides standardized mechanisms for creating partnership types. It is one of the APIs involved in an onboarding process.

- User Roles and Permissions API

This API is used to manage the user account with which the customer or staff can log-in and log-off the system and the permission for the user to enter or operate the special information.

- Entity Catalog Management API

This API provides the capability to manage the entity catalog API. The catalog entity item could be an entity, entity specification, product offering, service candidate, or resource candidate that appears in an entity catalog.

- Federated ID Management API

This API covers the operations required to allow an application (for instance a selfcare mobile app) request identity information about the individual that is making use of the functionality provided by such application (the user), or in general to allow an application to request identity related information about an individual to the system holding such identity information.

- Payment Management API

This API manages the payment action of the customer.

- Recommendation API

Recommendation API is used to recommend offering quickly based on the history and real-time context of customer. It is a real-time and personalized recommendation API. It is usually provided by e-commerce or BSS, CRM system in omni-channel.

- Customer Bill Management API

This API allows to find and retrieve one or several customer bills (also called invoices) produced for a customer. A customer bill is an electronic or paper document produced at the end of the billing process. The customer bill gathers and displays different items (applied customer billing rates generated during the rating and billing processes) to be charged to a customer. It represents a total amount due for all the products during the billing period and all significant information like dates, bill reference. The API model support the needs for the three basic billing types : postpaid periodical bill, postpaid real-time bill and prepaid real-time bill.

This API provides also operations to find and retrieve the details of applied customer billing rates presented on a customer bill.

And this API is also used to manage the creation request of a customer bill in real-time (on demand).

- Product Offering Qualification API

Product Offering Qualification API is one of Pre-Ordering Management API Family. Product Offering Qualification API goal is to provide Product Offering commercial eligibility. This API allows to

- request a specific productOffering eligibility
- look for productOffering eligibles (or not) from a product category
- look for productOffering eligibles (or not) from a productspec description.

- Service Problem Management API

This SPM API is used for the service providers (Defined as the Middle B) to manage the service problems in their service area.

- Resource Inventory API
The intent of this API is to provide a consistent/standardized mechanism to query and manipulate the Resource inventory.
- Alarm Management API
The Alarm Management API provides the standardized client interface to Alarm Management systems for creating, tracking and managing alarms among partners. The interface supports alarm management on both resources and services. The alarmed objects are not restricted to any particular technology or a vendor, so the API can be used in a wide variety of fault management cases.
- Resource Pool Management API
This Resource Pool management API provides feature of resource reservation at pre-order phase. In order to reserve physical product, logical product, and virtual product, the API uses the idea of resource pool.
- Party Interaction Management API
The Party Interaction Management API provides a mechanism to manage party interactions. Creation, update and retrieval.
- Geographic Address API

The Geographic Address Management API provides a standardized client interface to an Address management system.

- Geographic Site API

This API covers the operations to manage (create, read, delete) sites that can be associated to a customer, an account, a service delivery or other entities.

This API defines a Site as a convenience class that allows to easily refer to places important to other entities, where a geographic place is the entity that can answer the question “where?”, allowing to determine where things are in relation to the earth's surface, and can be represented either in a textual structured way (geographic address) or as a geometry referred to a spatial reference system (geographic location).

- Geographic Location API

A Geographic Location is a point, a surface or a volume defined by geographic point(s). These points should be associated with accuracy and a spatial reference.

The geographic location API provides a standardized client interface to a location management system.

- Communication API

It provides a standardized mechanism for Communication management such as creation, update, retrieval, deletion and notification of the system communication events.

Communication message means a notification approach in the format of a message which can be dispatched (sent) to the certain user by the system with the content which can be felt and understood by the recipient. The user can be either a final customer or a customer service agent. The message can reach the customer in different interaction channels, including: email, short message, mobile app notification (push).

Communication API performs the following operation on the resource of “Communication Message”. There are two types of operations provided in this API. One is the management of the request message body. Another is for sending the communication message to the customer.

- Usage Consumption API

The usage consumption API allow to view at a given point the balance and the consumption counters of the various buckets (SMS, Voice, Data for example) that one or more user(s) consume with each of his devices, according to the purchased offers and options.

A usage consumption report retrieves the data related to these balances and various consumption counters and calculated at the time of the request by the server.

So, the API allows to retrieve usage consumption report with information about balances and consumption counters for a given criteria: subscribed offer or option, a given device or a user for example.

- Performance Threshold API

The Performance Thresholding API provides a standardized client interface to Service and Resource Performance Management Systems for manipulating (create/update/delete) threshold/violation/exception rules. It enables alarms/notifications on exceptions and scheduling of threshold/violation/exception evaluation.

The API supports the ability to define a Threshold as a set of rules that determine when to raise an alarm and when to clear it for various severity levels. Correspondingly, for each rule used for raising alarms, the expected alarm fields of the Threshold Crossing Alarm (TCA) can be set over the interface. The API will support Performance Thresholds creation deletion and query.

- Shipment Tracking API

Telco companies often send goods to customers. Shipment information can be provided to them, so they can be aware of when things were shipped and when they will arrive. Typically, Telco's use different delivery companies to deliver things on their behalf and this API intends to abstract the end developers from that complexity by providing a single interface. A Shipment Tracking captures information about the current status of the shipment, the past checkpoints and the estimated arrival date. Via this API, tracking information can be retrieved by providing an order Id or the shipping company's tracking id.

3.2.2. Mapping between current TM Forum API and Information Framework

The mapping between the current TM Forum API and Information Framework ABE is described as the table below. The above figure is made based on the mapping results. The policy is that any API will be put in its main domain.

If one API covers several domains besides Customer and Engaged Party together, the main domain will be decided by the business. If one API covers Customer and Engaged Party together, the main domain will be decided by the business also but generally Engaged Party is prior to Customer.

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API L0 Domain
1.	Product Catalog Management API	Catalog	Product Domain ::Product Offering ABE :: Product catalog ABE	Product
		Category	To be investigated	
		Product Offering	Product Domain::Product Offering ABE	
		Product Specification	Product Domain::Product Specification ABE	
2.	Product Inventory Management API	Product	Product Domain::Product ABE	Product
3.	Billing Management API	Billing Account	Customer Domain::Customer ABE Engaged Party Domain::Additional Party Entities ABE	Customer
		Customer Bill Format	Customer Domain::Customer Bill ABE Engaged Party Domain::Party Revenue ABE	
		Customer Bill Presentation Media	Customer Domain::Customer Bill ABE Engaged Party Domain::Party Revenue ABE	
		Applied Billing Charge	Customer Domain::Applied Customer Billing Rate ABE Engaged Party Domain::Party Revenue ABE	
		Settlement Note Advice	To be investigated	
4.	Customer Management API	Customer	Customer Domain::Customer ABE	Customer
5.	Service Qualification API	Service Qualification	Customer Domain:: Customer Order ABE	Customer
6.	Quote Management API	Quote	Customer Domain:: Customer Interaction ABE	Customer
7.	Appointment API	Search Task	N/A	Customer
		Appointment	Common Domain::Calendar ABE	
8.	Prepay Balance Management API	Balance	To be investigated	Customer
		Balance Top Up	To be investigated	
		Bucket Balance	To be investigated	
		Accumulated Balance	To be investigated	
		Balance Adjustment	To be investigated	
		Balance Activity	To be investigated	
		Balance Reserve	To be investigated	
		Balance Unreserve	To be investigated	

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API L0 Domain
		Balance Deduct	To be investigated	
		Balance transfer	To be investigated	
9.	Product Ordering Management API	Product Order	Engaged Party Domain::Party Order ABE Customer Domain::Customer Order ABE	Customer & Engaged Party
10	SLA Management API	SLA Resource	Engaged Party Domain::Agreement ABE Customer Domain:: Customer Service Level Agreement ABE	Customer & Engaged Party
		SLA Violation Resource	N/A	
11	Service Quality Manage API	Service Level Objective	Service Domain:: Service Performance ABE	Service
		Service Level Specification	Service Domain:: Service Performance ABE	
12	Service Test Management API	Service Test	Service Domain::Service Test ABE	Service
		Service Test Specification	Service Domain::Service Test ABE	
13	Change Management API	Change Request	Engaged Party Domain:: Party Order ABE	Service
14	Service Catalog API	Service Catalog	Service Domain:: Service Specification ABE	Service
		Service Category	Service Domain:: Service Specification ABE	
		Service Candidate	Service Domain:: Service Specification ABE	
		Service Specification	Service Domain:: Service Specification ABE	
15	Service Inventory API	Service	Service Domain: Service ABE	Service
16	Service Problem Management API	Service Problem	Service Domain::Service Problem ABE	Service
17	Service Ordering API	Service Order	Service Domain::Service Order ABE	Service
18	Activation and Configuration API	Service	Service Domain::Service ABE	Service & Resource
		Resource(abstract)	Resource Domain::Resource ABE	
		Monitor	N/A	
19	Resource Ordering API	Resource Order	Resource Domain:: Resource Order ABE	Resource
20	Resource Catalog API	Resource Catalog	Resource Domain:: Resource Catalog ABE	Resource
		Resource Category	No existing ABE for mapping	
		Resource Candidate	Resource Domain:: Resource Specification ABE	
		Resource Specification	Resource Domain:: Resource Specification ABE	
21	Resource Function	Resource Function	Resource Domain:: Resource Specification ABE	Resource
		Resource Function/Heal	Resource Domain:: Resource ABE	

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API LO Domain
	Activation & Configuration API	Resource Function/Scale	Resource Domain:: Resource ABE	
		Resource Function/Migrate	Resource Domain:: Resource ABE	
22	Party Management API	Individual	Engaged Party Domain::Party ABE	Engaged Party
		Organization	Engaged Party Domain::Party ABE	
23	Privacy Management API	Party Privacy Profile	Engaged Party Domain::Party Privacy ABE	Engaged Party
		Party Privacy Profile Type	Engaged Party Domain::Party Privacy ABE	
		Party Privacy Agreement	Engaged Party Domain::Party Privacy ABE	
24	Onboarding Management API	Party Role	Engaged Party Domain:: Party ABE	Engaged Party
		Partnership Type	Engaged Party Domain:: Party ABE	
25	Partnership Type API	Partnership Type	Engaged Party Domain:: Party ABE	Engaged Party
26	Party Role API	Party Role	Engaged Party Domain:: Party ABE	Engaged Party
27	Agreement Management API	Agreement	Engaged Party Domain:: Agreement ABE	Engaged Party
		Agreement Specification	Engaged Party Domain:: Agreement ABE	
28	Geographic Address API	Geographic Address	Common Business Entities Domain::Location ABE	Common
		Geographic Address Validation	To be studied	
		Area	Common Business Entities Domain::Location ABE	
		Street	Common Business Entities Domain::Location ABE	
		Street Segment	Common Business Entities Domain::Location ABE	
29	Trouble Ticket Management API	Ticket Resource	Common Business Entities Domain::Trouble Ticket ABE	Common
30	Usage Management API	Usage Resource	Common Business Entities Domain::Usage ABE	Common
31	Performance Management API	Usage Specification Resource	Common Business Entities Domain::Usage ABE	(Note: Performance ABE is in Common Business Entity Domain in Information Framework)
		Measurement Production Job Resource	Common Business Entities Domain::Performance ABE	
		Measurement Collection Job Resource	Common Business Entities Domain::Performance ABE	
		Ad-hoc Collection Resource	Common Business Entities Domain::Performance ABE	
32	Loyalty Management	Loyalty Program Product Spec	Product Domain::Loyalty ABE	Product

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API L0 Domain
33	Promotion API	Promotion	Product Domain ::Product Offering ABE::Product Promotion ABE	Product
34	Shopping Cart API	Shopping Cart	Customer Domain::Customer Order ABE Engaged Party Domain :: Party Order ABE	Customer
35	User Roles & Permissions	UserRole	Common Business Entities:: User and Roles ABE	Common
		Permission	No existing ABE for mapping	
36	Entity Catalog Management	Entity Catalog	Common Business Entities Domain ::Catalog ABE :: Entity Catalog ABE	Common
		Category	No existing ABE for mapping	
		Entity Catalog Item	Common Business Entities Domain ::Catalog ABE :: Entity Catalog ABE	
		Entity Specification	Common Business Entities Domain :: Root Business Entities ABE	
		Association	Common Business Entities Domain :: Root Business Entities ABE	
		Association Specification	Common Business Entities Domain :: Root Business Entities ABE	
37	Document Management API	Document	Common Business Entities Domain :: Root Business Entities ABE No existing ABE for Document itself	Common
38	Payment Methods API	Voucher	Engaged Party Domain :: Party Revenue ABE :: Party Bill Collection ABE :: Party Payment ABE	Engaged Party
		Card Payment	Engaged Party Domain :: Party Revenue ABE :: Party Bill Collection ABE :: Party Payment ABE	
		Money Transfer	Engaged Party Domain :: Party Revenue ABE :: Party Bill Collection ABE :: Party Payment ABE	
39	Account Management API	Party Account	Engaged Party Domain:: Additional Party Entities ABE	Engaged Party
		Billing Account	Customer Domain::Customer ABE Engaged Party Domain::Additional Party Entities ABE	
		Settlement Account	Engaged Party Domain:: Party Revenue ABE :: Party Settlement ABE (preliminary)	
		Financial Account	Enterprise Domain :: Finance ABE	

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API LO Domain
		Billing Cycle Specification	Customer Domain::Customer Bill ABE Engaged Party Domain::Party Revenue ABE :: Party Bill ABE	
		Bill Format	Customer Domain::Customer Bill ABE Engaged Party Domain::Party Revenue ABE :: Party Bill ABE	
		Bill Presentation Media	Customer Domain::Customer Bill ABE Engaged Party Domain::Party Revenue ABE :: Party Bill ABE	
40	Federated ID API	User Info	To be studied	Customer
41	Alarm Management API	Alarm	Resource:: Resource Trouble ABE	Resource
42	Resource Inventory API	Logical Resource	Resource:: Resource ABE	Resource
		Physical Resource	Resource:: Resource ABE	
43	Geographic Location API	Geographic Location	Common Business Entities::Location ABE	Common
		Retrieve Geographic Location	No existing ABE for mapping	
		Retrieve Location Relation	No existing ABE for mapping	
44	Geographic Site API	Geographic Site		
45	Communication API	Communication Message	Common Business Entities:: Business Interaction ABE	Common
46	Usage Consumption API	Usage Consumption Report	Product Domain :: Product ABE :: Usage Volume Product ABE	Common
		Usage Consumption Report Request	N/A	
47	Performance Threshold API	Measurement Collection	Common Business Entities Domain :: Performance ABE	Common
		Measurement Production	Common Business Entities Domain :: Performance ABE	
		Ad-hoc Collection	Common Business Entities Domain :: Performance ABE	
48	Payment Management	Payment	Customer Domain :: Customer Bill Collection ABE::Customer Payment ABE Engaged Party Domain:: Party Bill Collection ABE::Party Payment ABE	Customer Engaged Party
		Refund	Customer Domain :: Customer Bill Collection ABE::Customer Payment ABE Engaged Party Domain:: Party Bill Collection ABE::Party Payment ABE	

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API L0 Domain
49	Product Offering Qualification	Product Offering Qualification	To be studied	Customer
50	Resource Pool Management API	Reservation	To be studied	Resource
		Resource Pool	To be studied	
51	Recommendation API	Recommendation	Product Domain::Product Offering ABE (partial)	Product
52	Shipment Tracking API	Shipment Tracking	To be studied	Common
53	Customer Bill API	Applied Customer Billing Rate	Customer Domain::Applied Customer Billing Rate ABE Engaged Party Domain::Party Revenue ABE	Customer
		Customer Bill	Customer Domain::Customer Bill ABE Engaged Party Domain::Party Revenue ABE	
		Customer Bill On Demand	No existing ABE for mapping	
54	Party Interaction	Party Interaction	Common Business Entities Domain :: Business Interaction ABE (partial)	Engaged Party

Note: Product Ordering API, SLA API, Activation and Configuration, Payment Management API appear in multiple domains due to the Information Framework mapping relationship.

3.2.3. Level 1 Map – Planned & Future API's

The following is a view of all the potential business APIs Level 1 Map that briefly describes each resource for the potential business API. In the years hereafter, those API definitions will be further explored and the efforts to build them are encouraged fully for all the industry.

Note: More details about the future API shall be supplemented when the API is confirmed. Currently the brief descriptions of those APIs are given in this chapter for understanding.

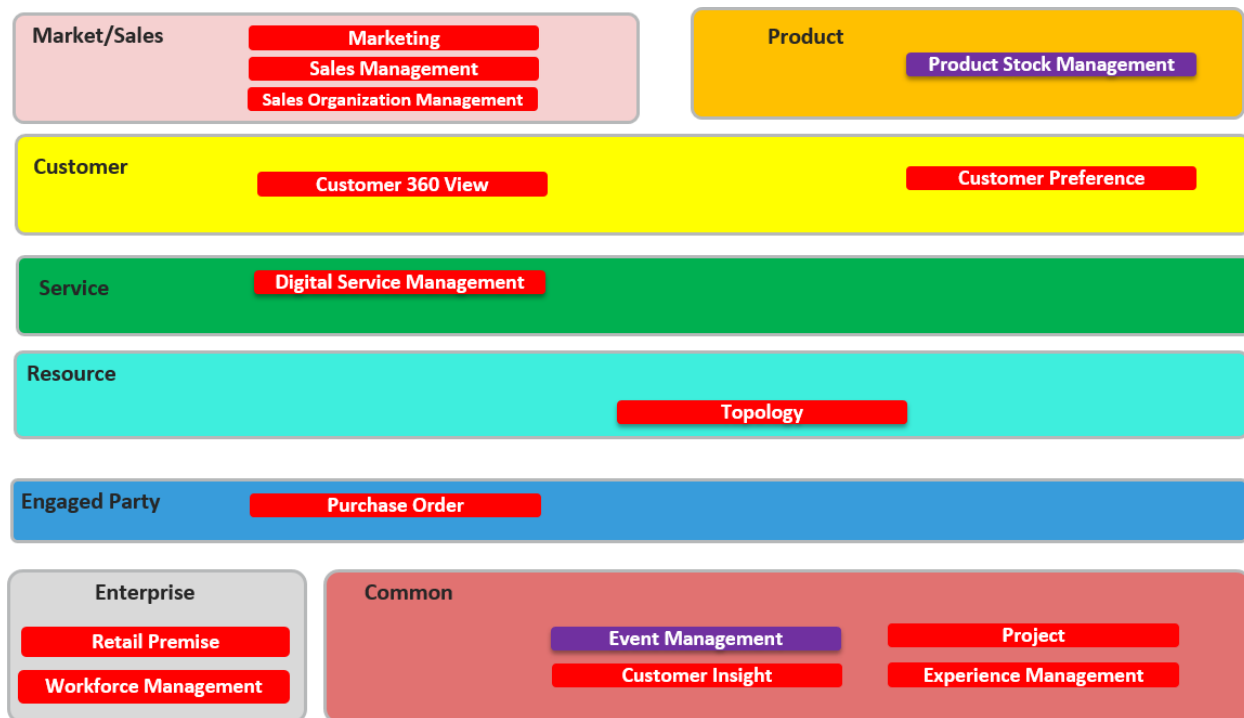


Figure 4 - API Map Level 1 (Planned & Future)

Here lists the APIs which have been planned in TM Forum and the proposed APIs which could be used to fill in the unfinished area in API Map.

This list is the guide for the API development in the future. The details of these APIs are expected to be found in the separate API specification.

Marketing/Sales

- Marketing API
This API is used to manage the campaign to represent the carrier-initiated marketing activity which aims at the better recognition about its brand and offerings by the market.
- Sales Management API
This API provides interfaces for Sales Lead, Sales Opportunity, Sales Quote and the other management capabilities to support the sales activities to build relationship with the prospect customer who could be a person or organization that has an interest in the goods and/or services and possibly become the actual customers with one or more Subscriptions.
- Sales Organization Management API
Provides the management capability for sales organization. The structure of a sales organization identifies how an organizations sales department is organized in order facilitate the organizations sales growth.

Product

- Product Stock Management API

This API checks availability of products in stock, and provides adjustment, reservation for the stock.

Customer

- Customer Preference API
It provides the capabilities to manage the customers' preference for their information access and control, including permission and profile.

- Customer 360 View
This API helps to have a full analysis of customer.

Service

- Digital Service Management API
Provides the management capabilities to manage the operating lifecycle of digital services – i.e. it exposes a simple “Operations API” defined in the “Digital Services Reference Architecture (DSRA)”. This includes the endpoints to manage the lifecycle of a digital service and their state and to manage alarms, SLAs, thresholds and related metrics in the context of application delivered and consumed through digital channels

Resource

- Topology API

This API is used to support application to have an up to date and accurate view of the topology and the vertexes and edges required to provide the service. At the same time, topology API could be used to feed other applications for value-adding – such as a layer network view for provisioning, or a vertical slice for assurance.

Enterprise

- Retail Premise API
This API is the management of the enterprise routine tasks, including project, purchase order and retail premises information enquiry.
- Workforce Management API
This API manages work force which supports and execute the manual work that can be sent to a workforce staff team to process.

Common

- Customer Insight API
This API is the statistics and business intelligence capabilities for the enterprise data.
- Project API
It is the management of internal enterprise project.
- Event Management

Event is any of the action triggered by the human or the system which requires the notification to the other system.

- Experience Management

Experience is the data and their impact which reflect the feeling of the user of the system. Based on the experience, the user can give judgment to the system about its usability.

3.2.4. Mapping between planned/future TM Forum API and Information Framework

The mapping between the planned and future TM Forum API and Information Framework ABE is described as the table below.

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API L0 Domain
1.	Marketing API	Marketing resource	Market/Sales Domain:: Marketing Campaign ABE	Market/Sales
2.	Sales Management API	Sales Lead	Market/Sales Domain:: Contact/Lead/Prospect ABE	Market/Sales
		Sales Opportunity	Market/Sales Domain:: Contact/Lead/Prospect ABE	
		Sales Task	N/A	
		Sales Account	Market/Sales Domain:: Contact/Lead/Prospect ABE	
		Sales Partner Account	Market/Sales Domain:: Contact/Lead/Prospect ABE	
3.	Sales Organization Management API	Sales Channel Management	Market/Sales Domain:: Sales Channel ABE	Market/Sales
4.	Product Stock Management API	Product Stock	No existing ABE for mapping	Product
		Product Stock Adjustment	No existing ABE for mapping	
		Product Stock Reservation	No existing ABE for mapping	
5.	Customer Preference API	Customer Preference	Customer::Customer ABE	Customer
		Customer Permission	No existing ABE for mapping	
6.	Customer 360 View API	Customer 360 View	Customer::Customer ABE	Customer
7.	Digital Service Management API	Digital Service	Service ::Service ABE	Service
8.	Topology API	N/A	Common Business Entities Domain :: Topology ABE	Resource
9.	Purchase Order API	Purchase Order	Engaged Party Domain :: Party Order ABE	EngagedParty
10.	Retail Premise API	Retail Premise	Enterprise Domain:: Enterprise Effectiveness ABE	Enterprise
11.	Workforce Management API	Work Order	Enterprise Domain ::Workforce ABE	Enterprise
12.	Project API	Project	Common Business Entities Domain:: Project ABE	Common
13.	Customer Insight API	Check Customer Credit Rating	Engaged Party Domain :: Additional Party Entities ABE	Common

#	API NAME(API)	Resource (API)	Information Framework Domain::L1 ABE-	API L0 Domain
		Customer Insight	Common Business Entities Domain:: Performance ABE Engaged Party Domain :: Party ABE :: Party Profile ABE	
14.	Event Management	NA	Common Business Entities Domain:: Event ABE	Common
15.	Experience Management	NA	No existing ABE for mapping	Common

3.3. Level 2 Map

Level 2 shows the API which inherits from the current TM Forum API or extends on the base of TM Forum API.

The legend in the Level 2 diagram is as:

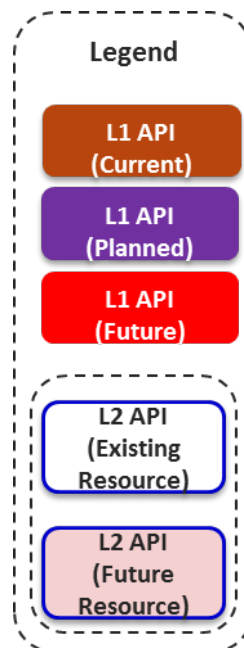


Figure 5 – Legend

3.3.1. API in Marketing/Sales



Figure 6- API in Marketing/Sales

The list of APIs in this portion is as follows:

- Marketing API
 - **Marketing Campaign** (as API resource)
Campaign represents the carrier-initiated marketing activity which aims at the better recognition about its brand and offerings by the market.
- Sales Management API
 - **Sales Lead** (as API resource)
A Sales Lead is the identification of a person or organization that has an interest in the goods and/or services provided in the prospect of them becoming Customers with one or more Subscriptions.
 - **Sales Opportunity** (as API resource)
Sales Opportunity is an 'opportunity' to generate revenue from a Sales Account or Sales Lead. Opportunities are the pending deals that need to be tracked and on which Sales Team plans and executes Sales Activities (Events and Tasks). Opportunities build the "pipeline" that will contribute to sales forecasts and planning.
 - **Sales Account** (as API resource)
The Sales Team within the operator or its partners opens a Sales Account for the potential customer on behalf of a Party. At this point the Party may be new to the Service Provider or the partner as a Sales Lead in which case they have the role of Prospect or may be an existing Customer. The Sales Account and associated Sales Opportunity form the basis around which sales activities are focused.
 - **Sales Partner Account** (as API resource)
It provides a capability for a consumer to search for and return the set of Sales Partner Account (the potential customer managed by the partner) records that match the supplied criteria in the request.
 - **Sales Task** (as API resource)
A Sales Task represents a business activity such as making a phone call or other to-do items. It can define and track activities for many different objects, including campaigns, accounts, contacts, and leads. An example of a Sales Task would be Manager scheduling a Task for an Assignee to make a call to the contact lead of a Sales Lead at a given time.
- Sales Organization Management API
 - **Sales Channel Management** (as API resource)
It provides the management capability for sales channels and organization which execute sales activities.

3.3.2. API in Product

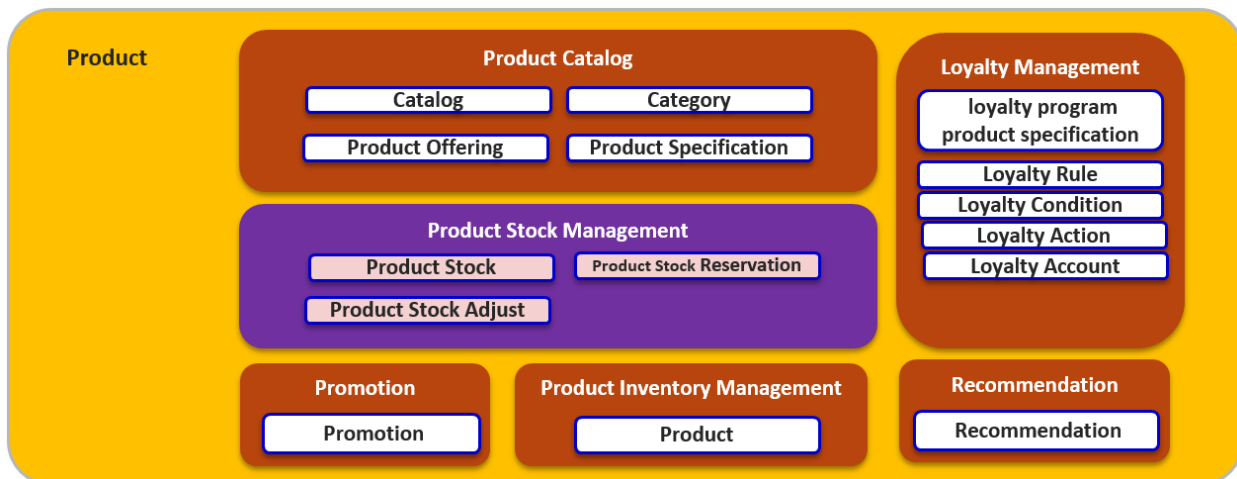


Figure 7 – API in Product

The list of APIs in this portion is as follows:

- Product Catalog API
 - **Catalog** (as API resource)

Catalog is a collection of Product Offerings, intended for a specific Distribution Channel, enhanced with additional information such as SLA parameters, invoicing and shipping detail.
 - **Category** (as API resource)

The category resource is used to group product offerings, service and resource candidates in logical containers. Categories can contain other categories and/or product offerings, resource or service candidates.
 - **Product Offering** (as API resource)

The Product Offering resource represents entities that are orderable from the provider of the catalog, this resource includes pricing information

Product Offering in a Product Catalog combines pricing and availability information with Product Specifications that describe the relationships between Products, the Services used to realize the Products and the Resources they require.
 - **Product Specification** (as API resource)

Product Specification Resource is a detailed description of a tangible or intangible object made available externally in the form of a Product Offering to customers or other parties playing a party role.
- Loyalty Management API
 - **Loyalty Program Product Specification** (as API resource)

A loyalty program product specification is a detailed description of a loyalty program made available externally in the form of a Loyalty Product to Loyalty Program Members. A loyalty program product specification defines one or more Loyalty Rules that have to be checked in order to identify the actions to apply
- Product Inventory Management API
 - **Product** (as API resource)

Product attributes are its identifier, name, description, status, related product offering, characteristics, specification, related billing account and parties, location, realized service and price information

- Promotion API

- **Promotion** (as API resource)

- Promotion is the online incentive which provides discount, gift or other substantial stimulation to encourage more consumption according to the purchased offering.

- Recommendation API

- **Recommendation** (as API resource)

- Recommendation is a type of automated (typically, conditional) system action to determine which products offerings to be presented as up-sells, cross-sells, and related products to each customer segment based on customer and session specific information. The design-time offers, and product recommendations may come from Marketing and Catalog. The run-time evaluation and presentment will be executed with contextual/session information. Normally, it supports the recommendation of the offer which is proper for the customer, provides Up-sell and cross-sell based on contextual and catalog rules.

- Product Stock Management API

- **Product Stock**(as API resource)

- The Product Stock service provides the capability for consumers to search for the stock levels and configuration of one or various Product Offering within Retail Premises (Store) or Storage Premises (Warehouse or Depot).

- The service supports the Inventory Management systems within Vodafone and forms part of the supply chain from Suppliers and Partners to get goods and services to a Point Of Sale for Customers involved in the selling processes.

- **Product Stock Adjustment**(as API resource)

- The Product Stock Adjustment service is an "Entity" service that provides a capability for consumers to sync stock levels in case a change is necessary in the Store Inventory Management (SIM) system. It will query the stock level at a particular location and adjust it, which includes notifying and synching the stock level with Retail or Storage (Warehouse or Depot) Premises.

- The service keeps the record of a change to the quantity of Product Stock for reasons other than stock receipt, stock dispatch or sale. Thus, Product Stock Adjustment may cover short shipment and shrinkage cases, like theft or wastage.

- **Product Stock Reservation**(as API resource)

- The Product Stock Reservation service provides the capability for consumers to reserve a defined quantity of stock (physical Product Offerings at a retail Premises or Warehouse) for a sales order and remove that quantity from the available stock.

3.3.3. API in Customer

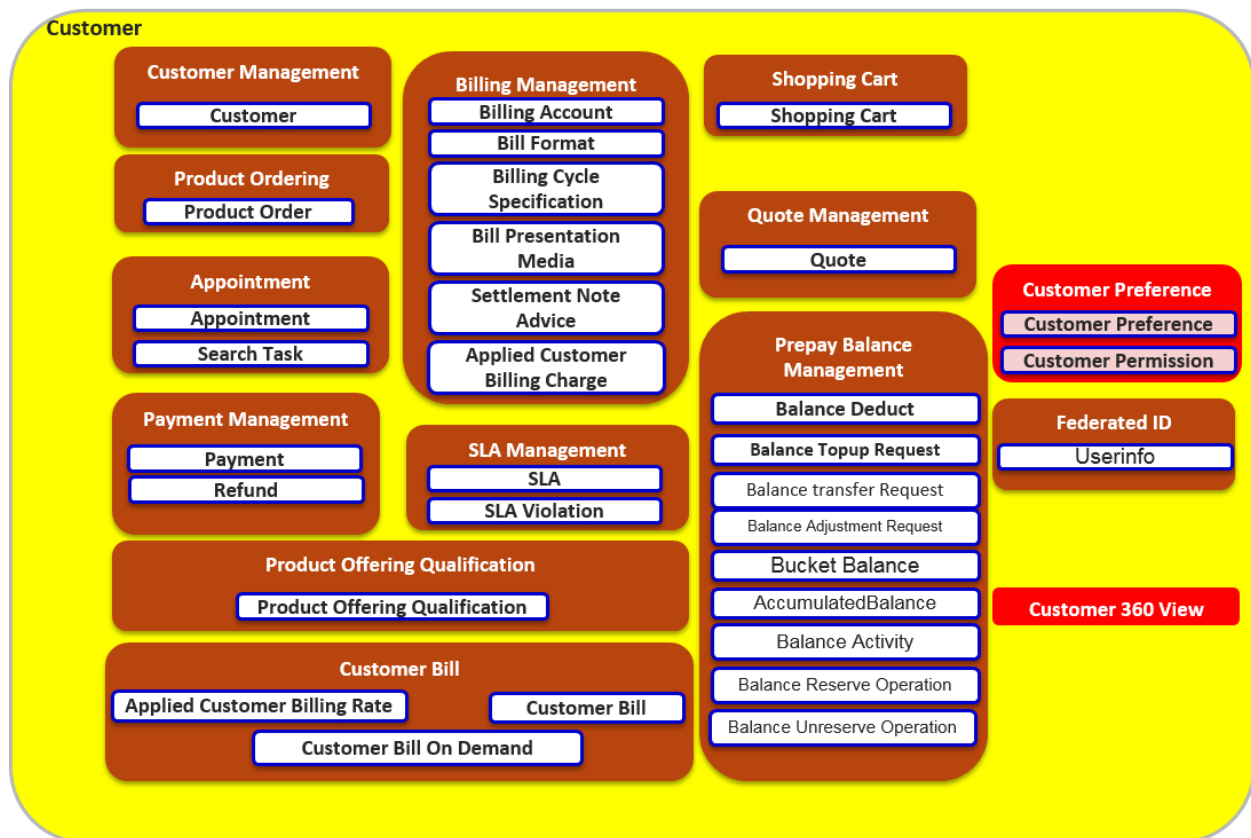


Figure 8 – API in Customer

Note: Product Ordering API, SLA API appear in multiple domains due to the Information Framework mapping relationship.

The list of APIs in this portion is as follows:

- Customer Management API
 - **Customer** (as API resource)
Customer represents a person or organization that buys products and services from the enterprise or receives free offers or services.
- Appointment API
 - **Appointment** (as API resource)
An appointment is a meeting with several persons, in one place, in order to do an action (an intervention, a sale ...). This action has a root, for example a trouble ticket.
 - **Search Task** (as API resource)
This task resource is used to look for free slots before booking an appointment (cf. operations).
- Product Ordering API
 - **Product Order** (as API resource)
It provides a standardized mechanism for placing a product order with all of the necessary order parameters. The API consists of a simple set of operations that interact with CRM/Order negotiation systems in a consistent manner. A product order is created based on a product offering that is defined in a catalog. The product offering identifies the product or

- set of products that are available to a customer, and includes characteristics such as pricing, product options and market.
- Product Offering Qualification API
 - **Product Offering qualification** (as API resource)

It checks whether the offering can be delivered to a specific location.

Service provider execute Product-Offering Qualification task to get the customer location

Feasibility include Commercial and Technical eligibility.
- Payment Management
 - **Payment** (as API resource)

The Payment resource represents a performed payment. It contains both information about the payment and the payment method used to perform it. That information could be used for reconciliation purposes with payment gateways, to centralize information of several payment gateways in a single place, to keep track of payment methods used in case refunds must be made, etc.
 - **Refund** (as API resource)

Operations over refund resource are expected to be performed by a privileged user/system (such as an admin)
- Customer Bill API
 - **Customer Bill**(as API resource)

A billing account is a detailed description.

The customer bill is linked to one or more documents that can be downloaded via a provided url.
 - **Applied Customer Billing Rate**

A customer bill displays applied billing rates created before or during the billing process
 - **Customer Bill On Demand**

This resource is used to manage the creation request of a customer bill in real-time (on demand).
- Shopping Cart API
 - **Shopping Cart** (as API resource)

The shopping cart is widely used for the temporarily selection and reservation of offerings in e-commerce and retail purchase.
- Quote Management API
 - **Quote** (as API resource)

It provides a mechanism for placing a customer quote based on a product offer in a catalog

It identifies available product to a customer w/ pricing, product options and agreement
- Prepay Balance Management API
 - **Bucket Balance** (as API resource)

The Bucket Balance resource represents and tracks the amount remained or owed for a certain type of service by certain customer. The balance is associated to a specific product or reference to a product (i.e.: a commercial id such as an msisdn) and a service type .This resource covers the main attributes defined in class CustomerAccountBalance defined in SID (validFor, remainedAmount).
 - **Accumulated Balance**(as API resource)

The Accumulated Balance resource represents and tracks the aggregated amount remained or owed in certain account which is owned by certain customer for a set of buckets. This is an optional resource that may be useful to allow a consumer understand the specific balance for

a given service/product, aggregating all the buckets that hold balance for that service (e.g.: acquired by the customer, offered for free as a promotion by the operator, granted by another customer,) without requesting one by one the balance in all the buckets and then perform the addition.

- **Balance Top up Request (as API resource)**

The resource is a detailed description of a recharge operation requested over a subscription.

- **Balance Transfer Request (as API resource)**

The resource is a detailed description of credit transfer operation requested between two subscriptions.

- **Balance Adjustment Request (as API resource)**

The resource is a detailed description of credit adjustment operation performed on a given subscriptions.

- **Balance Activity(as API resource)**

The Balance Activity resource is a detailed description of a specific balance-related action that has happened over a given bucket balance. Typically, a recharge/transfer/adjustment request creates one activity, but a request for an auto-top-up operation actually triggers multiple periodic balance-related activities.

- **Balance Reserve Operation(as API resource)**

The Balance Reserve Operation resource is a detailed description of a balance reserve operation requested over a bucket (defined by a specific product or reference to a product (i.e.: a commercial id such as an msisdn) and a service type)

- **Balance Unreserve Operation(as API resource)**

The Balance Unreserve Operation resource is a detailed description of a balance unreserve operation requested over a product

- **Balance Deduct(as API resource)**

The Balance Deduct task resource is a detailed description of deduction operation. If balanceReserve Resource ID is contained in the deduct request message, the reserved balance will be performed deduct operation (if part of the reserved balance is deducted, the remaining amount will be released); if balanceReserve Resource ID is not contained in the deduct request message, the balance will be deducted directly.

- **Customer Preference API**

- **Customer Permission (as API resource)**

The Customer Permission service is an "Entity" service that provides capabilities to store and manage the set of Permissions configured by a Party in the Role of Customer.

- **Customer Preference (as API resource)**

The Customer Preference service is an "Entity" service that provides capabilities to store and manage the set of Preferences configured by a Party in the Role of Customer.

- **Customer 360 View API**

The customer 360° view interface allows finding and retrieving the main information regarding the customer. A customer view gathers different information to be presented to a user or the customer himself. This interface is designed in order to be consumed by any kind of front end which need to present a customer view.

In order to build a customer 360°View, the following information can be retrieved:

- ◆ Individual or organization information,
 - ◆ Contact medium: mail, telephone, ...,
 - ◆ Product order,
 - ◆ Product : offers subscribed by the customer,
 - ◆ Customer problem,
 - ◆ Interaction : inbound and outbound contact,
 - ◆ Customer bill,
 - ◆ Usage
- Bill Management API
 - Billing Account (as API resource)
A billing account is a detailed description of a customer's bill structure.
 - Customer Bill Format (as API resource)
A detailed description of the way in which a customer's bill is presented.
 - Customer Billing Cycle Specification (as API resource)
It is a detailed description of when to initiate a billing cycle and the various sub steps of a billing cycle.
 - Customer Bill Presentation Media (as API resource)
It is a means of communicating a Customer Bill, supported by the associated bill format. For example, post mail, email, and web page.
 - Applied Customer Billing Charge (as API resource)
It includes the bill amount, usually of money, for which a person or an organization is financially liable.
 - Settlement Note Advice (as API resource)
The settlement is about transferring money receiving by a partner to another partner. The settlement is notified to the partner with a settlement note advice containing details in settlement lines.
 - SLA Management API
 - Service Level Agreement (SLA) (as API resource)
The SLA API provides a standardized interface for SLA life cycle Management (SLA Negotiation, SLA configuration, SLA Activation/enforcement, SLA Operations, SLA reporting) between a Customer and a Service Provider which provides offers (product with attached SLA in its catalog) the customer can discover, browse, trigger and order.
 - **SLA Violation** (as API resource)
It is about SLA violation / consequence handling.
 - Federated ID API
 - **Userinfo** (as API resource)
The Userinfo resource represents a class that allows to define identity related information of an individual (i.e.: a Party).

3.3.4. API in Service

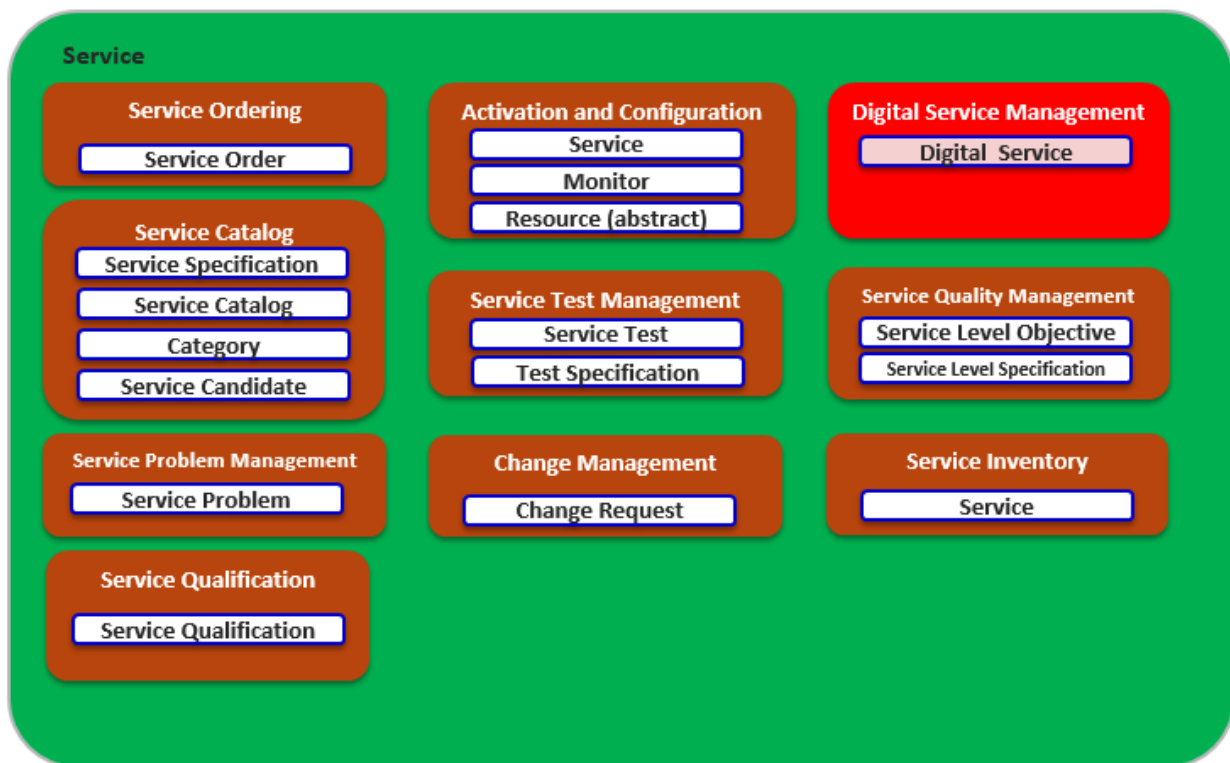


Figure 9 – API in Service

Note: Activation and Configuration API appears in multiple domains due to the Information Framework mapping relationship.

The list of APIs in this portion is as follows:

- Activation and configuration API
 - **Service** (as API resource)

Services may be the process to support the accomplishment of sales or customer request, normally it requires human handled work. It can be activated or configured by the Activation and Configuration API.
 - **Monitor** (as API resource)

This is to monitor the activation progress.
 - **Resource (abstract)** (as API resource)

It is to describe the common set of attributes shared by all concrete resource (e.g. cross-connect, shelf). The abstract Resource collection is a container or collection for all the “concrete” resources.
- Service Ordering API
 - **Service Order** (as API resource)

Service order composed of order lines (order Items):

 - Line “1”: Ordering of a new simple Service that needs a physical delivery place and an appointment to be delivered
 - Line “2”: Modification of a characteristic value of an already owned Service, and change the user associated to this Service

- Line “3”: Ordering of a new simple Service that needs (is supported by) another already owned Service

- Service Catalog API
 - Service Catalog (as API resource)

The root entity for catalog management.

- **Service Category** (as API resource)

The category resource is used to group service candidates in logical containers.

- Service Candidate (as API resource)

A Service Candidate is an entity that makes a Service Specification available to a catalog. A Service Candidate and its associated Service Specification may be published - made visible - in any number of Service Catalogs, or in none.

- Service Specification (as API resource)

Service Specification is a class for representing a generic means for implementing a particular type of Service.

- Service Inventory API
 - **Service** (as API resource)

Service is an abstract base class for defining the Service hierarchy. All Services are characterized as either being possibly visible and usable by a Customer or not. This gives rise to the two subclasses of Service: Customer Facing Service and Resource Facing Service.

- Service Problem Management API
 - Service Problem (as API resource)

Service problem is generated based on the information declared by Middle B or the event information notified from infrastructure providers (Defined as the First B) who provide the infrastructure of cloud or network.
- Service Qualification Management API
 - Service Qualification (as API resource)

Service Qualification API is used to perform a technical eligibility. It allows to retrieve a list of services that are technically available in the context of the interaction (place, party, service characteristics).
- Digital Service Management API
 - Digital Service (as API resource)

This digital service is a kind of service that managed in this API.

- Service Test Management API
 - **Service Test** (as API resource)

A service test specification describes the service test in terms of parameters to be configured and measures to be taken.

- Service Test Specification (as API resource)

A service test is an entity that exists that exists for a controlled test invocation on a service. The service test is executed according to a schedule and contains service test configuration parameters that are to be applied at execution time, and service test measures that result.

- Change Management API
 - **Change Request** (as API resource)

Change Management process is to respond to the customer's changing business requirements while maximizing value and reducing incidents, disruption and network. The Change Management API provides the standard integration capabilities between external applications and Change Management Application. The API consists of a simple set of operations that interact with Change Request in a consistent manner. A Change Request is an IT service management discipline. The objective of change management in this context is to ensure that standardized methods and procedures are used for efficient and prompt handling of all changes to control IT infrastructure and Network, in order to minimize the number and impact of any related incidents upon service.

- Service Quality Management API
 - **Service Level Objective** (as API resource)

Service level objective is the quality goal for a Service Level Specification defined in terms of parameters and metrics, thresholds, and tolerances associated with the parameters.

- **Service Level Specification** (as API resource)

A service level specification is a pre-defined or negotiated set of Service Level Objectives, and consequences that occur, if the objectives are not met.

3.3.5. API in Resource

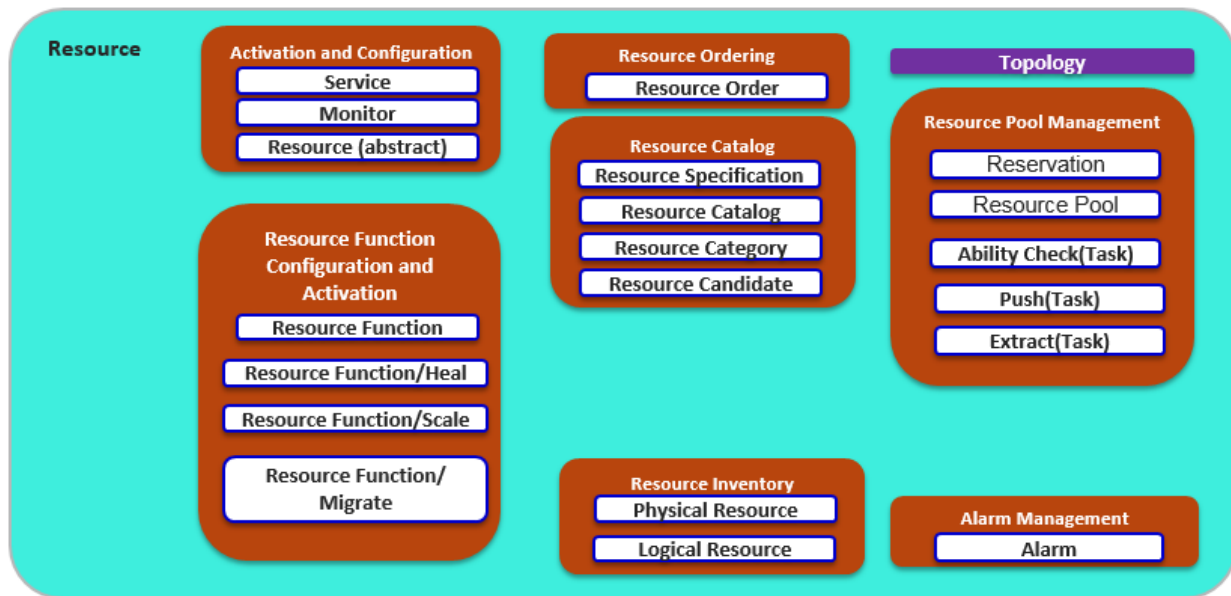


Figure 10 – API in Resource

Note: Activation and Configuration API appears in multiple domains due to the Information Framework mapping relationship.

The list of APIs in this portion is as follows:

- Activation and configuration API
 - Note: the description for this API is same as the API in “Service” domain
- Resource Function Activation & Configuration API
 - Resource Function (as API resource)

A network service that is composed from one or many virtual or physical network functions.
 - Resource Function /Heal (as API resource)

Task resource used to request heal of the network service
 - Resource Function /Scale (as API resource)

Task resource used to request scale of the network service
 - Resource Function /Migrate (as API resource)

Task resource used to request migration of the network service

- Resource Catalog API
 - Resource Catalog (as API resource)

The root entity for catalog management.

- Resource Category (as API resource)

The category resource is used to group service candidates in logical containers.

- Resource Candidate (as API resource)

A Resource Candidate: Is an entity that makes a Resource Specification available to a catalog.

- Resource Specification (as API resource)

A Resource Specification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a Resource Specification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular Resource Specification.

- Resource Ordering API
 - **Resource Order** (as API resource)

A Resource order composed of 3 order lines (orderItems):

- Line "1": Ordering of a new simple Resource that needs a physical delivery place and an appointment to be delivered.
 - Line "2": Modification of a characteristic value of an already owned Resource, and change the user associated to this Resource.
 - Line "3": Ordering of a new simple Resource that needs (is supported by) another already owned Resource (in this case for example a new Resource that relies on the owned mobile access Resource to be installed).
- Alarm Management API
 - **Alarm** (as API resource)
The alarm contains information about a given alarm condition of an alarmed Managed Object.
 - Resource Inventory API
 - **Logical Resource** (as API resource)
The Logical Resource is an "Entity" providing capabilities to update and retrieve sets of logical resources, for example, mobile number is one type of logical resource
 - **Physical Resource** (as API resource)
The Physical Resource is an "Entity" providing capabilities to update and retrieve sets of physical resources allocated to the Service Provider by external 3rd party providers.
 - Topology API
The resource of this API is not decided in this version.
 - Resource Pool Management API
 - **Resource Pool Resource**
Resource Pool can indicate total capacity, free capacity and used capacity of a resource and it takes ownership to assign resource.
 - **Reservation Resource**
It is used to reserve resources or resource capacity to be assigned / allocated from First B Pool according to request from Middle B.
 - **Push TASK**
Increases the size of the Pool by adding resources to the Middle B assignment Pool according to the request of Middle B.
 - **Extract TASK**
To reduce the size of the Pool by extracting resources of the Middle B assignment Pool according to the request of Middle B.

3.3.6. API in Engaged Party

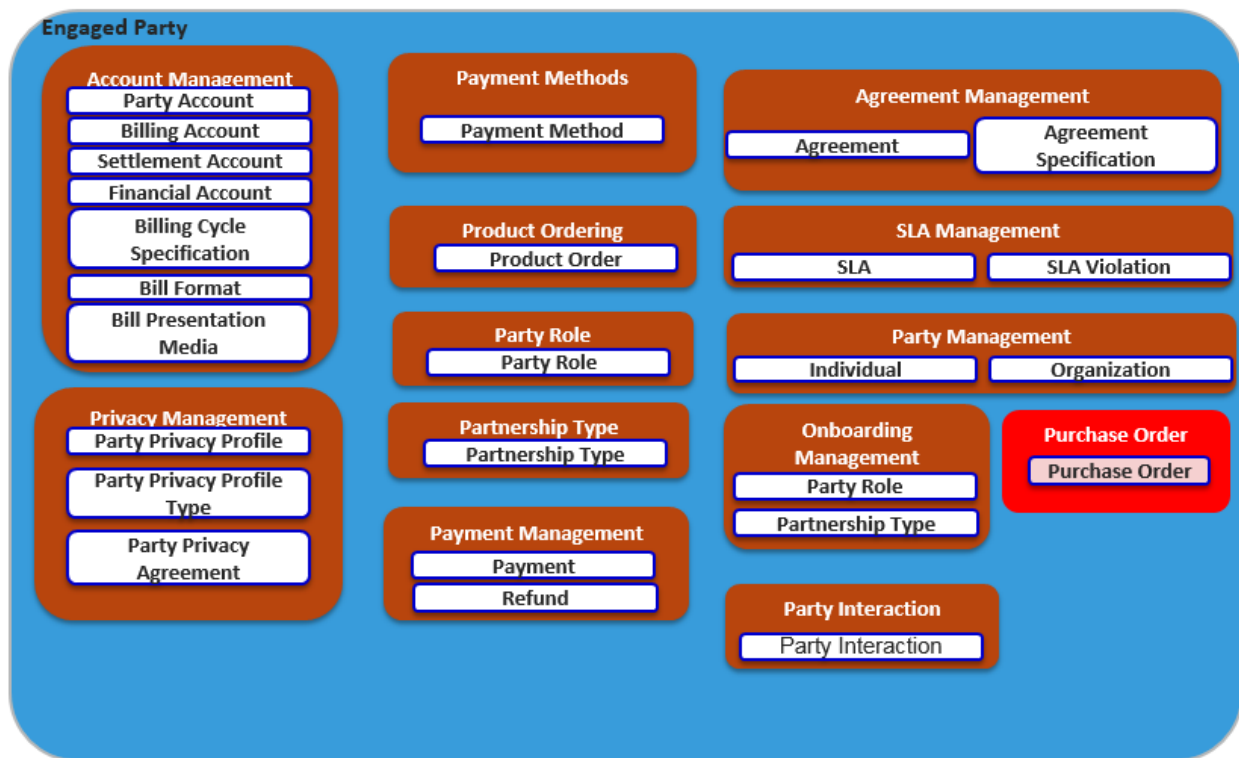


Figure 11 – API in Engaged Party

Note: Product Ordering API, SLA API appear in multiple domains due to the Information Framework mapping relationship.

The list of APIs in this portion is as follows:

- Party Interaction API
 - Party Interaction (as API resource)

The party interaction resource is used for indicating an interaction within a user and a company (e.g.: visit to a store, call to a call center, message received in a marketing campaign, etc.). A user interaction might result in other actions, such as submitted orders or raised tickets. A user interaction has a description, a reason, an originator (the user or the company) and contains a reference to an account, to other related parties and to the channel where it occurred.
- Account Management API
 - Billing Account (as API resource)

A billing account is a detailed description of a customer's bill structure.
 - **Bill Format** (as API resource)

A detailed description of the way in which a customer's bill is presented.
 - Billing Cycle Specification (as API resource)

It is a detailed description of when to initiate a billing cycle and the various sub steps of a billing cycle.
 - Bill Presentation Media (as API resource)

It is a means of communicating a Customer Bill, supported by the associated bill format. For example, post mail, email, and web page.
 - Party Account

Account used for billing or for settlement purposes concerning a given party (an organization or an individual).

- **Settlement Account**

A party account used for settlement purposes. It includes a description of the structure used for the settlement (frequency, presentation media, format and so on).

- **Financial Account**

An account of money owed by a party to another entity in exchange for goods or services that have been delivered or used. A financial (account receivable account/account payable) aggregates the amounts of one or more party accounts (billing or settlement) owned by a given party.

- Privacy Management API

- **Party Privacy Profile** (as API resource)

A Party Privacy Profile represents the privacy choices made by a Party Role.

- **Party Privacy Profile Type** (as API resource)

A Party Privacy Profile Type represents a type description for Party Privacy Profiles.

- **Party Privacy Agreement** (as API resource)

A Party Privacy Agreement represents the approval made by the Party about a Party Privacy Profile.

- Payment Methods API

- **Payment Method** (as API resource)

The Payment Method resource represents an instantiated payment method that can be of different types. Some fields are common to all types while the method specification details depend on the type.

- Party Management API

- **Individual** (as API resource)

Individual represents a single human being (a man, woman or child). The individual can be a customer, an employee or any other person that the organization needs to store information about

- **Organization** (as API resource)

Organization represents a group of people identified by shared interests or purpose. Examples include business, department, and enterprise.

- Product Ordering API

- **Product Order** (as API resource)

Note: the description is same as this API in "Customer" domain

- SLA Management API

Note: the description is same as this API in "Customer" domain

- Party Role

- **Party Role** (as API resource)

Party role is the basis data for the customer and partner. It is the business incarnation of the party.

This resource represents the party role, the part played by a party in a given context.

- Partnership Type

- **Partnership Type** (as API resource)

A partnership type contains all the information for the setup of a partnership of a given kind. This includes the list of identified role types for the partnership with the corresponding agreement specifications.

- Onboarding Management API Family

- **Party Role** (as API resource)

- This resource represents the party role, the part played by a party in a given context.

- **Partnership Type** (as API resource)

- A partnership type contains all the information for the setup of a partnership of a given kind. This includes the list of identified role types for the partnership with the corresponding agreement specifications.

Note:

Onboarding Management API has strong dependencies with the following management APIs:

- **Party Management API:** used to query, create, update or delete information on *individuals* or *organizations* that will be onboarded;
- **Agreement Management API:** used to query, create, update or delete agreements and agreement specifications. These agreements need to be created and updated when signed by the involved parties.
- **Billing Management API** (It will be enriched and renamed as Account Management API in the future): used to retrieve, create, update or delete different kinds of accounts that made be needed in the context of the onboarding process, such as billing or settlement accounts and financial accounts.

Other indirect APIs dependencies when using this API are:

- **Product Catalog Management API:** used to connect agreements to product offerings
- **Product Inventory Management API:** retrieval of products related to product offerings.
- **Product ordering management API:** establishing order on available products.

- Agreement Management API

- **Agreement** (as API resource)

- An agreement represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other business entities, such as products, services, and resources and/or their specifications.

- **Agreement Specification** (as API resource)

- A template of an agreement that can be used when establishing partnerships.

- Purchase Order API

- **Purchase Order** (as API resource)

- It supports capability to manage purchase order raised by the operator to 3rd party suppliers.

3.3.7. API in Enterprise



Figure 12 – API in Enterprise

The list of API in the Enterprise domain is as follows:

- Retail Premise API
 - **Retail Premise** (as API resource)
It is the information maintenance of physical retail outlets.
- Workforce Management API
 - **Work order** (as API resource)
This API manages work force which supports and execute the manual work that can be sent to a workforce staff team to process.

3.3.8. API in Common

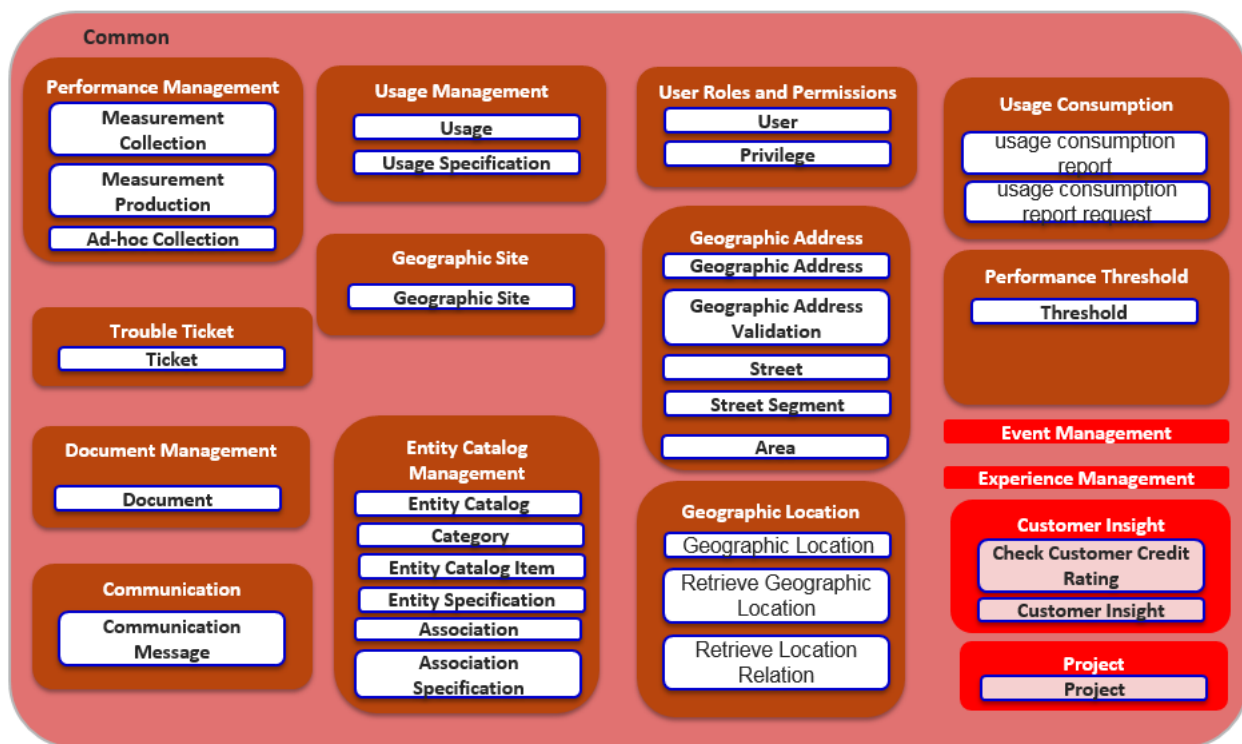


Figure 13 – API in Common

The list of API in the Common domain is as follows:

- Performance Management API
 - **Measurement Collection** (as API resource)
Measurement Collection Job is used to control the periodic collection of performance indicators, implemented as a sub-entity of the PM Measurement Job.
 - **Measurement Production** (as API resource)
Performance measurements production functionality is about controlling the generation of indicators.
 - **Ad-hoc Collection** (as API resource)
Ad-hoc collection is an on-demand collection method of Performance measurements results from a consuming application, requesting spontaneously a set of data for a set of network or service testing resources for a given temporal context.
- Trouble Ticket API
 - **Ticket** (as API resource)
It represents a problem which affects the customer experience. It can be raised by the customer (a complaint) or by the service provider.
- Document Management API
 - **Document** (as API resource)
It provides the operations to synchronize documents and document versions across systems. It also provides operations for uploading documents by Users as well as for viewing of documents online.
- Communication API
 - **Communication Message** (as API resource)

Communication message means a notification approach in the format of a message which can be dispatched (sent) to the certain user by the system with the content which can be felt and understood by the recipient. The user can be either a final customer or a customer service agent. The message can reach the customer in different interaction channels, including: email, short message, mobile app notification (push).

- Customer Insight API
 - Customer Insight (as API resource)
It supports capability to retrieve customer analytics profile.
 - Check Customer Credit Rating (as API resource)
Credit rating is to manage the estimation of credit score of the customer.
- Geographic Location API
 - Geographic Location (as API resource)
Structured textual way of describing a geographic location. A geographic location allows describing through coordinate(s), a point, a line or a space.
 - Retrieve Location (as API resource)
This resource is used to manage the request and response to retrieve a thing (any object).
 - Retrieve Location Relation (as API resource)
This resource is used to manage the request and response to retrieve the relation between two geographic locations.
- Geographic Address API
 - Geographic Address (as API resource)
Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently).
 - Geographic Address Validation (as API resource)
This resource is used to manage geographic address validation request and response.
 - **Area** (as API resource)
Area corresponds to a geographic area as a city, a locality, a district, etc.
 - **Street** (as API resource)
A street is a road in a city or locality that has buildings that are usually close together along one or both sides.
 - **Street Segment** (as API resource)
Street Segment corresponds to a part of a given street referenced by a number (22) or a set of number (22-24). Sometimes a suffix is added.
This level can represent an entire address, which is accessible through hyperlinking.
- Geographic Site API
 - Geographic Site (as API resource)
The Geographic Site resource represents a class that allows to easily refer to Places important to other entities (such as a customer, an account, a product,).
- Entity Catalog Management API
 - **Entity Catalog** (as API resource)
The root entity for entity catalog management. An entity catalog can be defined as a collection of entity catalog items and arranging them in a particular manner based on the need.

- **Category** (as API resource)
The (entity) category resource is used to group entity catalog items in logical containers. Categories can contain other categories.
- **Entity Catalog Item** (as API resource)
An EntityCatalogItem is an entity that allows any ProductOffering, ServiceCandidate, ResourceCandidate, or even any object inheriting from RootEntity, to be available to a catalog. The CatalogSpecification and Policy of the catalog governs the content of the EntityCatalogItem. This API addresses only EntitySpecification items extended from RootEntity as EntityCatalogItem.
- **Entity Specification** (as API resource)
EntitySpecification is a class that offers characteristics to describe a type of entity. Entities are a generic construct that may be used to describe bespoke business entities that are not effectively covered by the existing SID model.
- **Association** (as API resource)
Association is the class that describe a relationship between two or more entity specifications based on a given association specification.
- **Association Specification** (as API resource)
Association Specification is an association class that describes a type of relationship between two entities. This is a generic construct that may be used to describe relationship types and roles. The role and type of each entity in the relationship is given by an association role specification. Two role specifications are embedded in the association specification
- Usage Management API
 - **Usage** (as API resource)
An occurrence of employing a Product, Service, or Resource for its intended purpose, which is of interest to the business and can have charges applied to it. It is comprised of characteristics, which represent attributes of usage.
 - **Usage Specification** (as API resource)
A detailed description of a usage event that are of interest to the business and can have charges applied to it. It is comprised of characteristics, which define all attributes known for a particular type of usage.
- User Roles and Permissions API
 - **User** (as API resource)
This API is used to manage the user account with which the customer or staff can log-in and log-off the system.
 - **Privilege** (as API resource)
Privilege means the permission for the user to enter or operate the special information. This API is used to manage the privilege to control the data visibility and operation scope for the user.
- Performance Threshold API
 - **Threshold** (as API resource)
Performance Threshold Management is about identifying exceptional behavior of monitored performance indicators. Threshold Management is built as an additional layer on top of standard Performance Monitoring. A PM Producing application that is already producing indicators is asked to set a Threshold over a Performance Indicator and raise an alarm when this Threshold is crossed. When the indicator returns to its normal behavior an alarm clear alarm event is sent.

- Project API
 - **Project** (as API resource)
It is the management of internal enterprise project.
- Usage Consumption
 - Usage Consumption Report (as API resource)
A usage consumption report enables to know at a given point the balances and the consumption counters related to various buckets (SMS, Voice, Data for example). It could be calculated for a device identified by a public identifier (msisdn number for a mobile device for example or PSTN or VOIP number for a fix device), for a subscribed offer or option or for a user.
 - Usage Consumption Report Request (as API resource)
This resource is used to manage the calculation request of a usage consumption report.
- Event Management
The resource of this API is not decided in this version.
- Experience Management
The resource of this API is not decided in this version.

4. API Dependency Relation

In TM Forum API, some of them depend on the other. Such dependency relationship exists and impacts the adaptation of API in the digital environment. The recognition of those dependencies is important and meaningful for the users to understand the priority and sequence of the API.

Here the API dependencies are listed for the published ones due to the consideration of maturity and stability of API.

The dependency can be categorized as “internal” and “external”.

- ✓ Internal dependency means the resource entities inside the API have dependency between them, such as “Catalog” resource of “Product Catalog Management API” depends on “Category” resource of the same API
- ✓ External dependency means the resource entities of the API have dependency upon the resource of another API

Normally, external dependency should be paid more attention.

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
Product Catalog Management API	Catalog	Y	Category	Y	Party Management API	Party
	Category	N		N		
	Product Offering	Y	Category Product Specification	Y	Service Catalogue API	Service Candidate
		Y	Product Specification		SLA Management API	SLA
					Resource Catalog API	Resource Candidate
	Product Specification	N		Y	Party Management API	Party
					Service Catalog	Service Specification
					Resource Catalog	Resource Specification
Product Inventory	Product	N		Y	Product Catalog	Product Specification

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
Management API					Product Catalog Management API	Product Offering
					Ressource Inventory API	Ressource
					Service Inventory API	Service
					Account API	BillingAccount
					Agreement Management API	Agreement
					Party Management API	Party
Shopping Cart API	Shopping Cart	N		Y	Product Catalog API	Product Offering
				Y	Product Catalog	Product Specification
				Y	Party Management API	Party
Account Management API	Party Account	Y	Financial Account	Y	Party Management API	Party
		Y	Bill Presentation Media	Y	Onboarding Management API	Party Role
		Y	Billing Cycle Specification			
		Y	Bill Format			
	Billing Account	Y	Party Account			
	Settlement Account	Y	Party Account	Y	Onboarding Management API	Party Role
	Financial Account	N		Y	Onboarding Management API	Party Role
				Y	Party Management API	Party
	Billing Cycle Account	N		N		
	Billing Cycle Specificatiion	N		N		

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
	Bill Format	N		N		
	Bill Presentation Media	N		N		
Billing Management API	Billing Account	Y	customer bill presentation	Y	Party Management API	Party
			Customer billing cycle specification		Customer Mgmt	Customer Account
			Customer bill Format		Customer Mgmt	Payment Mean
	Customer Billing Cycle Specification	N		N		
	Customer Bill Format	N		N		
	Customer Bill Presentation Media	N		N		
	Applied Billing Charge	N		N		
	Customer Account					
	Settlement Note Advice	N		Y	Product Catalog	Product Specification
Customer Management API	Customer	Y	Customer Account	Y	Party Management API	Party
Service Qualification API	Service Qualification	N		N	Service Inventory API	Service
					Service Catalog	Service Specification
					Address API	Address
	Product offering Qualification	N		Y	Product Catalog,	Product Offering
					Product Inventory API,	Product
					Address API	Address
Quote Management API	Quote	N		Y	Product Catalog	Product Offering
					Document Management	Attachment
					Customer Mgmt	Customer

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
					Product Inventory API	Product
					Billing Management API	BillingAccount
					Appointment API	Appointment
					Agreement Management API	Agreement
					Party Management API	Party
Appointment API	Search Task (available slot)	N		Y	Party Management API	Party
	Appointment	N		Y	Document Management	Attachment
				Y	Party Management API	Party
Prepay Balance Management API	Balance	N		Y	Party Management API	Party
					Product Inventory	Product
	Balance Adjustments	N		N		
	Balance Topup Request	N		Y	Party Management API	Party
					Product Inventory	Product
					Customer Mgmt	Payment Mean
	Balance transfer	N		Y	Party Management API	Party
					Product Inventory	Product
Product Ordering Management API	Product Order	N		Y	Party Management API	Party
					Billing Management API	Billing Account

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
					Product Inventory	Product
					Appointment API	Appointment
					Product Catalog	Product Offering
SLA Management API	SLA (Service Level Agreement)	N		Y	Party Management API	Party
	SLA Violation Resource	Y	SLA	Y	Party Management API	Party
Service Quality Manage API	Service Level Objective	N		N		
	Service Level Specification	Y	Service Level Objective	N		
Service Test Management API	Service Test	N	Service Test Specification	Y	Service Inventory	Service
	Service Test Specification	N		Y	Service Catalog	Service Specification
Change Management API	Change Request	N		Y	Party Mgmt API	Party
		N		Y	SLA Management	SLA
Service Catalog API	Service Catalog	Y	Service Category	Y	Party Management API	Party
	Service Category	N		N		
	Service Candidate	Y	Service Category	Y	SLA	SLA
		Y	Service Specification			
	Service Specification	N		N		
Service Inventory API	Service	N		Y	Service Catalog	Service Specification
				Y	Service Ordering	Service Order
				Y	Party Management API	Party
Service Problem Management API	Service Problem	N		Y	Party Management API	Party
				Y	Alarm API	Alarm

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
				Y	Trouble Ticket	Trouble Ticket
				Y	Resource Inventory	Resource
				Y	SLA	SLA Violation
				Y	Service Inventory	Service
Service Ordering API	Service Order	N		Y	Service Inventory	Service
				Y	Service Catalog	Service Specification
				Y	Service Catalog	Service Catalog
					Party management API	Party
					Appointment management	Appointment
Activation and Configuration API	Service	N		N		
	Resource(abst ract)	N		N		
	Monitor	N		N		
Resource Ordering API	Resource Order	N		Y	Resource Inventory	Resource
					Party Management API	Party
					Resource Catalog	Resource Specification
					Appointment management	Appointment
Resource Catalog API	Resource Catalog	Y	Service Category	Y	Party Management API	Party
	Resource Category	N		N		
	Resource Candidate	Y	Resource Category	Y	SLA	SLA
		Y	Resource Specification			
	Resource Specification	N		N		
Resource Function	Network Service	N		Y	Service Catalogue API	Service Specification

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
Configuration & Activation API					Service Inventory API	Service
					Party Management API	Party
					Resource Inventory API	Resource
	VNF	N		N		
	PNF	N		N		
	Network Service Heal or VNF/Heal	N		N		
	Network Service Scale or VNF/Scale	N		N		
	Network Service Migrate or VNF/Migrate	N		N		
Party Management API	Individual	N		N		
	Organization	N		N		
Privacy API	Party Privacy Profile	Y	Party Privacy Profile Type	Y	Product Catalog	Product Offering
	Party Privacy Profile Type	N		N		
	Party Privacy Agreement	Y	Party Privacy Profile	Y	Agreement	Agreement
Onboarding Management API	Party Role	N		Y	Party Management API	Party
				Y	Agreement	Agreement
				Y	Customer Mgmt	Payment Mean
	Partnership Type	N		Y	Agreement Management	Agreement Specification
Agreement Management API	Agreement	Y	Agreement Specification	Y	Product Catalog	Product Offering
				Y	Onboarding Management	Party Role
	Agreement Specification	N		Y	Party Management API	Party
					Service Catalog	Service Category

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
Geographic Address API	Geographic Address	N		N		
	Geographic Address Validation					
	Area	N		N		
	Street	Y	Address	N		
	Street Segment	Y	Address	N		
Trouble Ticket API	Ticket Resource	N		Y	Party Management API	Party
Usage Management API	Usage Resource	Y	Usage Specification	Y	Party Management API	Party
					product inventory API	product
Performance Management API	Measurement Production Job Resource	N		N		
	Measurement Collection Job Resource	N		N		
	Ad-hoc Collection Resource	N		N		
Loyalty Management API	Loyalty Program Product Spec	N		Y	Product Catalog	Product Specification
API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
Promotion API	Promotion	N				
Recommendation API	Recommendation	N				
				Y	Shopping Cart API	Shopping Cart
				Y	Product Catalog	Product Offering
				Y	Product Ordering	Product Order
				Y	Party Management	Party

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
				Y	Entity Catalog	Category
Communication API	Communication Message	N		Y	Party Management	Party
Document Management API	Document	N		Y	Party Management	Party
Usage Consumption API	Usage Consumption report	N		Y	Party Management	Party
	Usage Consumption report Request	N		Y	Product Management	Product
Geographic Site API	Geographic Site	N		Y	Party Management	Party
					Geographic Address	Geographic Address
					Geographic Location	Geographic Location
Geographic Location API	Geographic Location	N		N		
Product Offering Qualification API	Product Offering Qualification	N		Y	Party Management	Party
					Product Catalog	Product Category
					Product Catalog	Product Offering
					Product Catalog	Product Specification
Entity Catalog Management API	Entity Catalog	Y	Category	Y	Party Management	Party
	Entity Category	Y	Entity Catalog Item	Y	Party Management	Party
	Entity Catalog Item	Y	Category	N		
			Entity Specification	N		
	Entity Specification	N		Y	Party Management	Party
Performance Threshold API	Performance Threshold	N		N		
Alarm Management API	Alarm	N		N		
Federated ID API	Userinfo	N		Y	Party Management	Party

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
					Party Role	Party Role
Resource Inventory API	Logical Resource	N		Y	Resource Catalog	Resource Specification
					Party Management	Party
	Physical Resource	N		Y	Resource Catalog	Resource Specification
					Party Management	Party
Resource Pool Management API	Reservation Item	Y	Resource Pool			
				Y	Resource Inventory	Resource
				Y	Party Management	Party
				Y	Product Catalog	Product Offering
	Resource Pool	N		Y	Activation & Configuration API	Resource collection
Payment Methods API	Payment Method	N		Y	Party Management API	Party
Payment Management API	Payment	N		Y	Party Management API	Party
					Account Management	Account
Product Ordering API	Product Order	N		Y	Party Management API	Party
					Payment Management	Payment
					Product Catalog	Product Offering
					Product Catalog	Product Specification
					Appointment Management	Appointment
Party Interaction API	Party Interaction	N		Y	Party Management API	Party
UserRoles and Permissions API	permissions	N		N		
	User Role	N		N		

API NAME(API)	Resource (API)	Internal Dependency Exists or Not	Internal Resource Entity Depended	External Dependency Exists or Not	External API Depended	External API Resource Depended
Customer Bill API	Bill	N		Y	Account Management	Financial Account
				Y	Account Management	Billing Account
				Y	Party Management	Party
				Y	Payment Management	Payment
	Applied Customer Billing	Y	Bill	N		
	Customer Bill on Demand	N		Y	Party Management	Party
Shipment Tracking API	Shipment Tracking	N		Y	Geographic Location	Geographic Location
Party Role API	Party Role	N		Y	Party Management API	Party
					Agreement	Agreement
					Customer Mgmt	Payment Mean
Partnership Type API	Partnership Type	N		Y	Agreement Management	Agreement Specification

5. API Map Usage examples

This chapter includes some practical examples of platforms as different service providers have defined them and how Open API Map is used to enable a platform-based business model as defined in the Digital Platform Reference Architecture (DPRA).

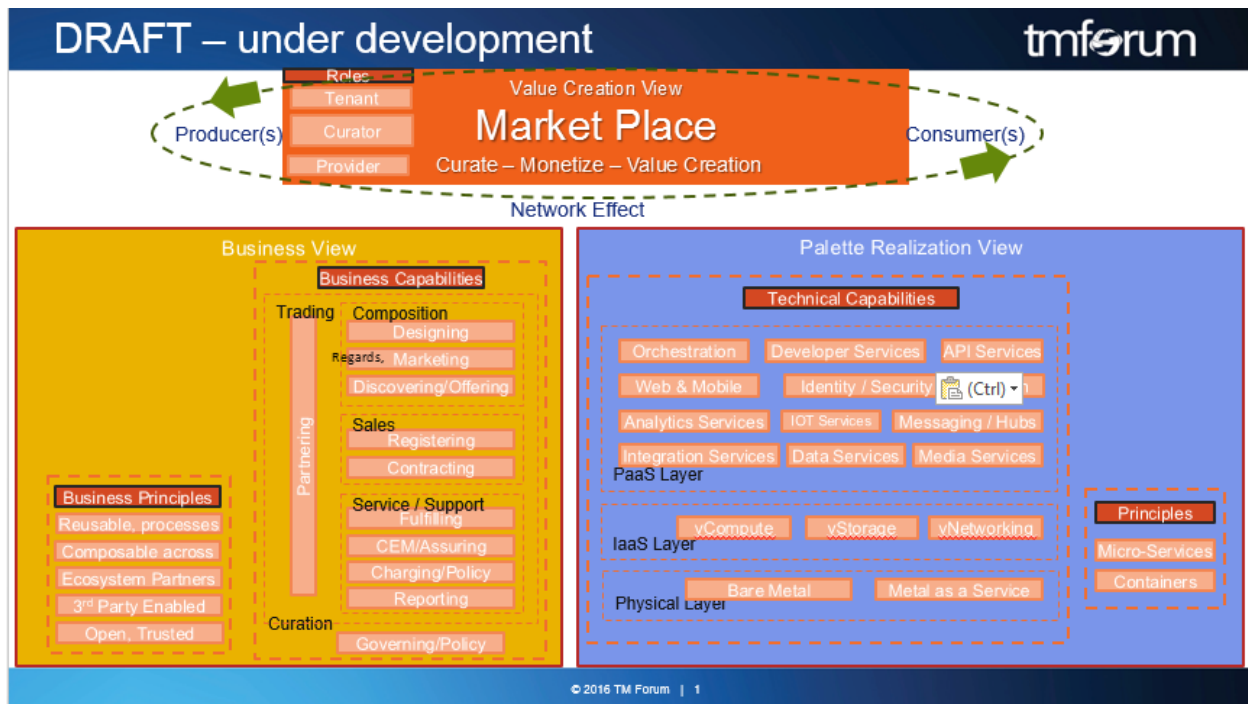


Figure 14 – DPRA Platform template

The value of transformation to a platform model to service providers as explained in the DPRA is that the platform model can help:

- To unbundle ownership of the infrastructure (e.g. network access or cloud infrastructure) from the offering of value-added services to consumers
- Promotes disaggregation of network functions (e.g.) which enable the offering of rich and new value-added composite services
- Allow for the aggregation of service elements from multiple producers to offer new and value-added services (i.e. service composition across producer boundaries).

Platform based business models can also be used to efficiently leverage an operator's core assets and provide competitive differentiation.

The service providers are already in the middle of many service chains so have a natural placement to easily become the curator between producers and consumers of services. They must transform from a "Producer" mentality to a "Curator" or match maker mentality.

The DPRA is based on 3 key concepts:

- Platforms: Instances are units of deployment defined by enterprises using templates defined by Standards Defining Organizations (SDO), including the TM Forum.

They are 'Black box like', can be Vendor /Operator specific, allow for internal evolution and innovation, and whose implementation can be agile and flexible i.e. time varying.

Platform instances are not standardized by SDOs but by enterprises.

- Platform Capabilities are exposed by a Platform and are the units of integration used to create composed capabilities from multiple Platform Capabilities / Platforms.

Platform Capabilities define the rules and best practices for using the combination of Open APIs used to realize them.

Platform Capabilities can be standardized by SDOs.

- Open APIs which are the units of interoperability that realize Platform Capabilities and can be conformance and interoperability tested.

Open APIs are standardized and governed by the TM Forum as well as other SDOs.

Digital Platform Reference Architecture shows the relationship between platforms and API's.

Informal Metamodel

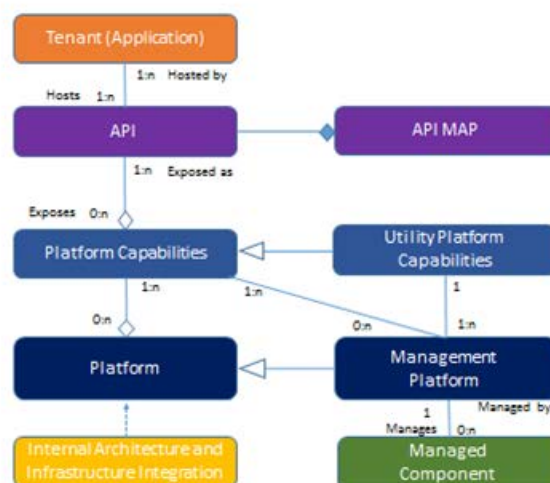


Figure 15 – Platform meta-model

A Platform is typically made up of a number of systems managed as a coherent unit to deliver a well-defined set of business capabilities through simple to use Open APIs.

5.1. API Deployment View in Vodafone

As seen above, The TMF Open API's help realize a set of Platform capabilities that can be standardized by SDO's.

At Vodafone, the Open APIs are shown in context of Platform instances, a deployment view of the Open API's.

A deployment view to the API Map:

- Enables to focus on platform capabilities that need to be exposed through API's rather than on how to implement them using different systems/applications
- Enables inter-operability whilst enabling innovation in the implementation. While consuming platform capabilities and integrate through Open API's we should not have any knowledge of the underlying systems or processes that deliver those capabilities.

The deployment view is the overall illustration of the API's in the API Map where we show each API in context of the platform that acts as "system of record" or "authoritative master" for the associated API resources.

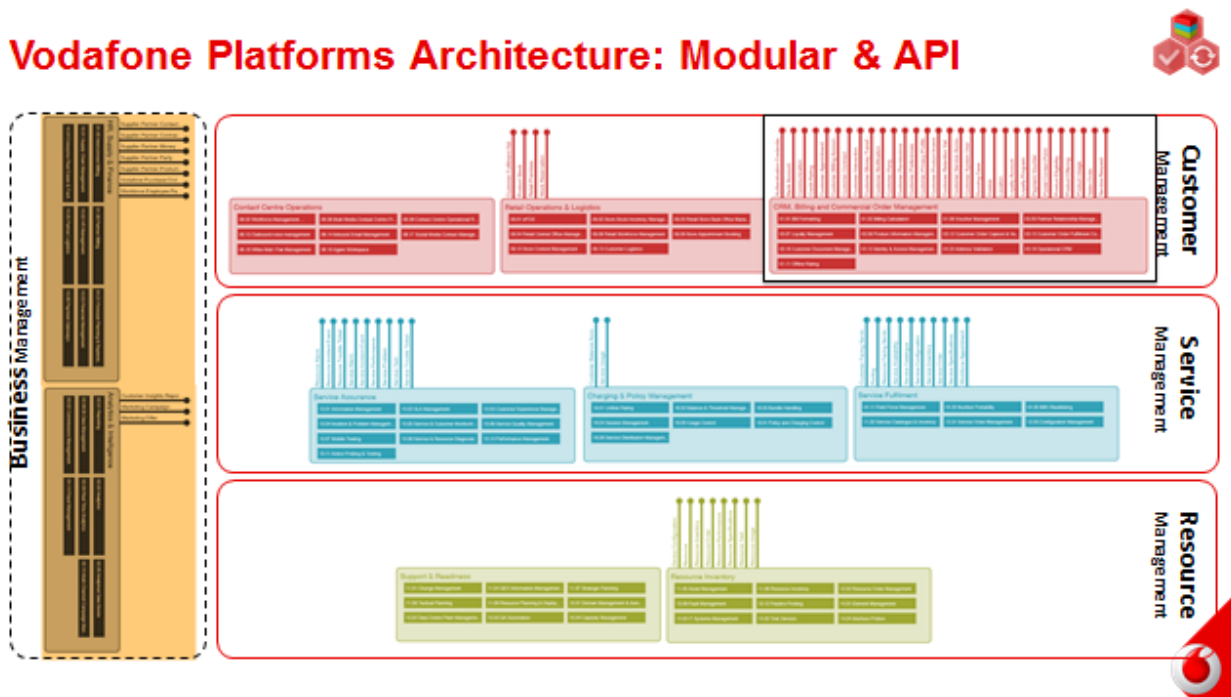


Figure 16 – Vodafone Platform Architecture

A number of platform instances have been established in Vodafone which:

- can be implemented differently in different company
- But expose the same business capabilities and realised using the same APIs
- Enable agility across the companies by re-use of the same APIs and platform capabilities.

In Vodafone Platform Architecture:

- Customer Management layer includes CRM, Billing and Order Management platform, Retail Operations and Logistics, Contact Centre Operations platform
- Service Management layer includes Service fulfilment, Service Assurance, Charging & Policy Management platforms
- Resource Management layer includes Resource Inventory & Lifecycle Management, Resource activation & Configuration, Resource Fault & Performance platforms

The 'Deployment view' of the API Map is organized in multiple levels:

- Level 0: the platform layers, 4 business abstraction layers: customer & product management, service management, resource management, business management
- Level 1: The main TM Forum APIs, typically the primary resource
- Level 2: the component resources exposed within the API

5.1.1. Level 0 of API Deployment View

The deployment view of the API's is aligned with the Customer/Service/Resource view as the overall business abstractions as below:

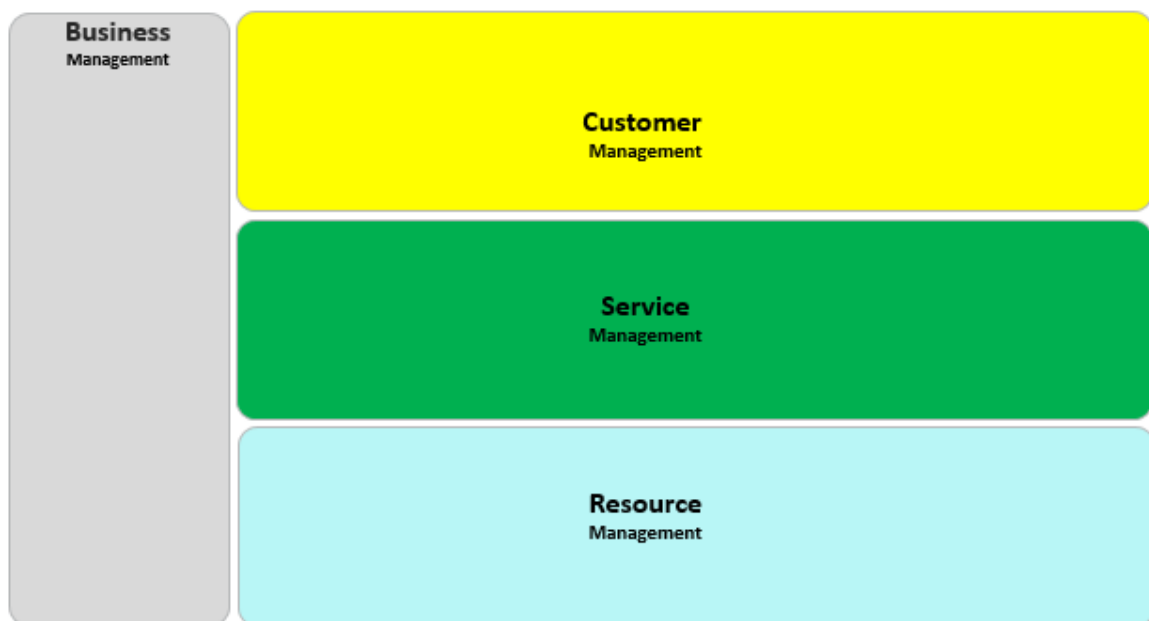


Figure 17 – Level 0 Deployment view

5.1.2. Level 1 of API Deployment View

In this Level 1 depiction, the API' from the API Map are mapped to the platform layers above
The Level 1 of API Deployment view is as the following:



Figure 18 – Level 1 Deployment view

Here are the current APIs in the deployment view:



Figure 19 – Level 1 Deployment view (Current APIs)

The APIs in the planned or future status can be mapped on the deployment view as the following:

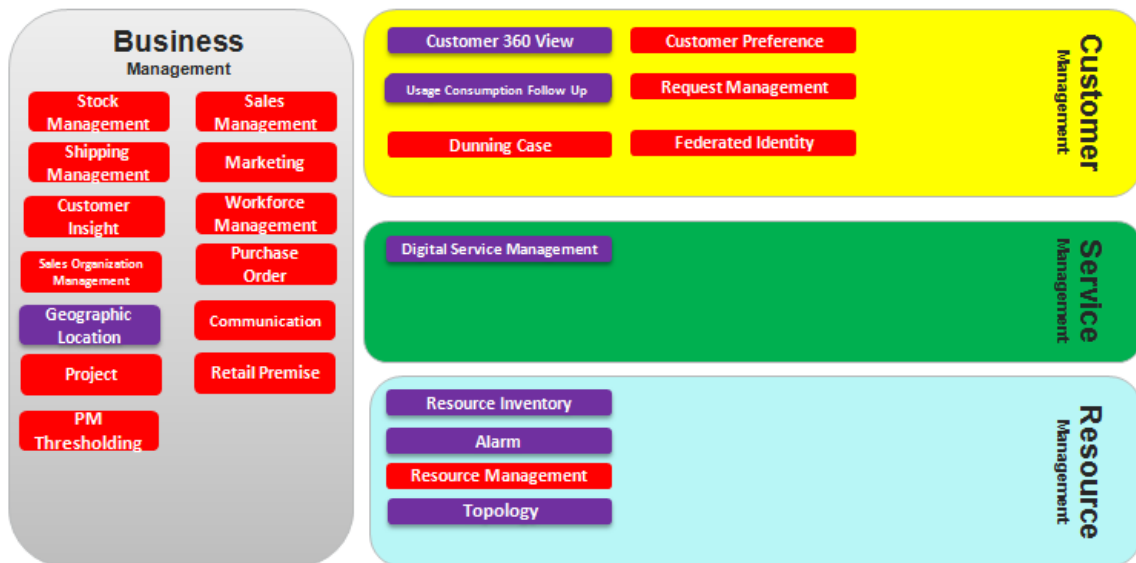


Figure 20 – Level1 Deployment view (Planned & Future API's)

5.2. API Deployment View in BT

BT's business functionality has been partitioned into a set of cooperating IT platforms that enables:

- Reusable common capabilities (SDK's/API's) – keeping engineering costs down
- Reusable process blocks – consistent customer experience

BT has 26 platforms and 700 systems which drives simplicity and ruthless standardization and is intended to minimise whole life costs, reduce cycle time for launching new capabilities and facilitate business agility.

BT's Platform Architecture

26 platforms

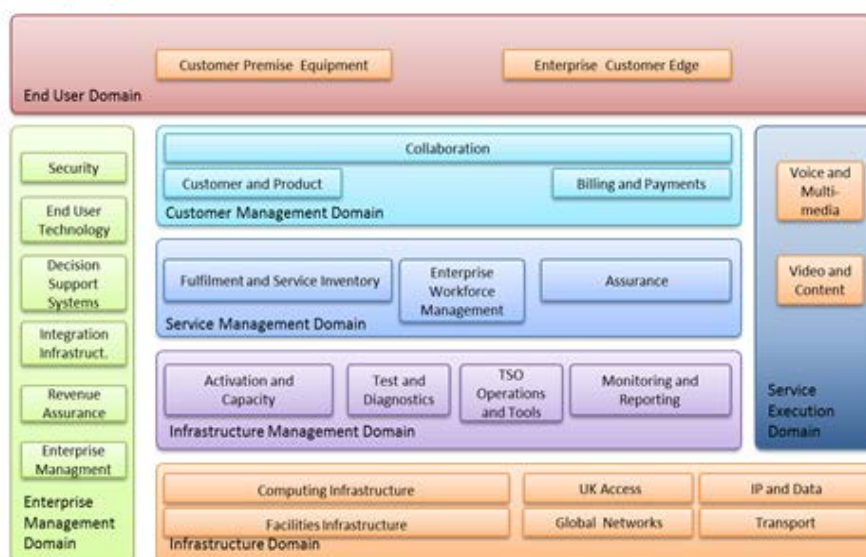


Figure 21 – BT Platform Architecture

5.2.1. Level 0 of API Deployment View

The deployment view of the API's is aligned with the Customer/Service/Resource view as the overall business abstractions as below:

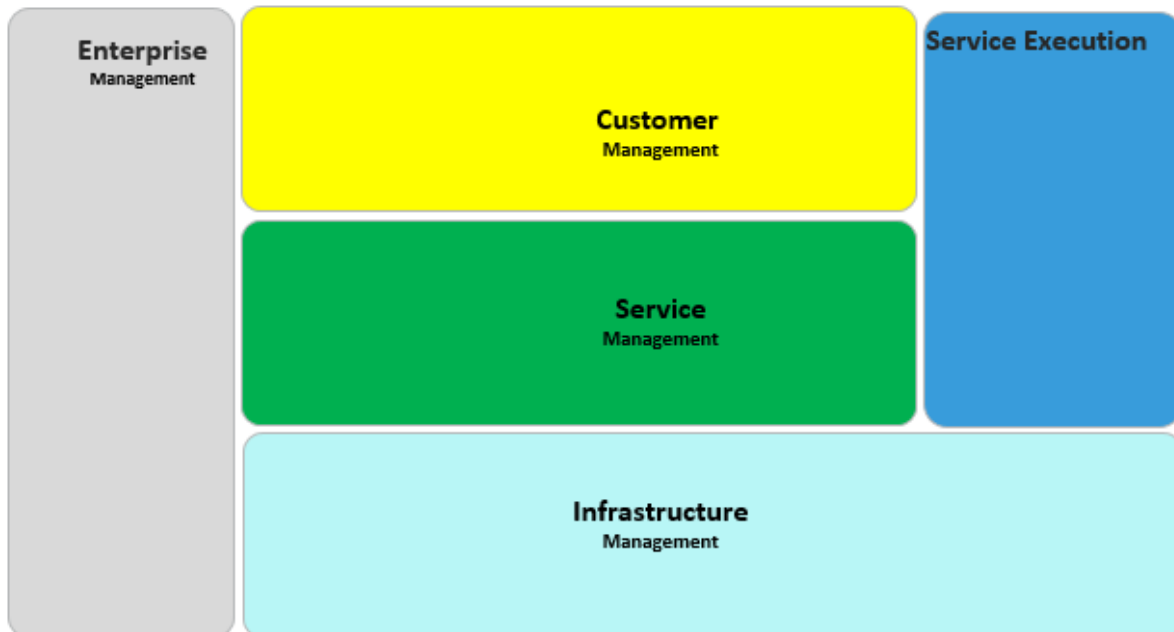


Figure 22 – Level 0 BT API Deployment view

5.2.2. Level 1 of API Deployment View

In this Level 1 depicted, the API' from the API Map are mapped to the platform layers above
The Level 1 of API Deployment view is as the following:



Figure 23 – Level 1 BT Deployment view

Here are the current APIs in the deployment view:



Figure 24 – Level 1 BT Deployment view (Current API's)

The APIs in the planned or future status can be mapped on the deployment view as the following:

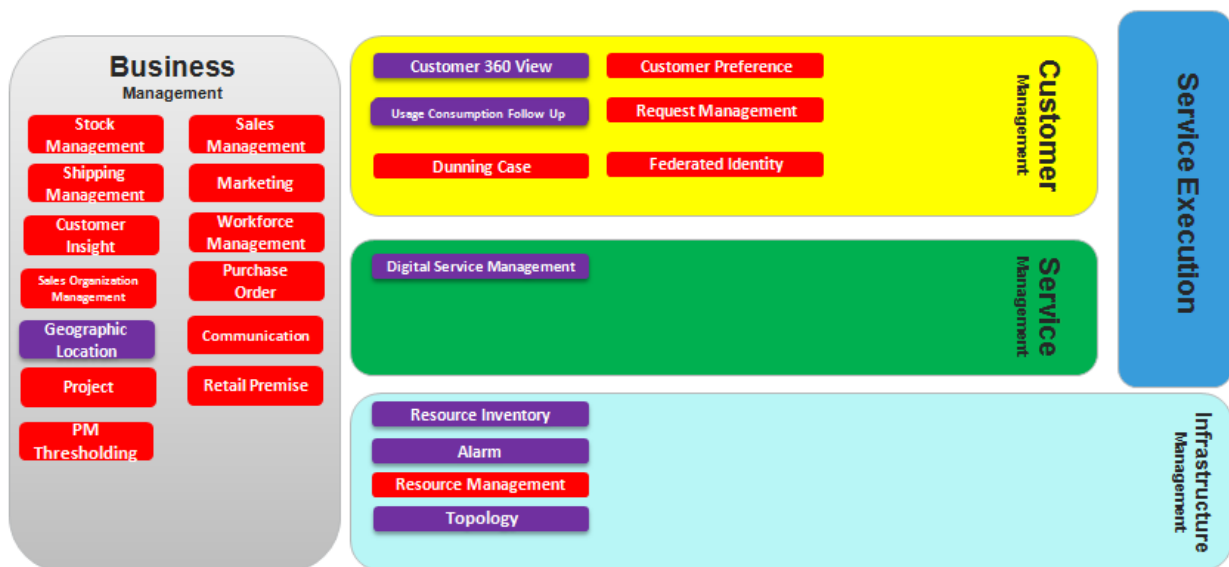


Figure 25 – Level 1 BT Deployment view (Planned & Current)

5.3. API Deployment View in Orange

Orange's business functionality has also been partitioned into a set of cooperating IT platforms depicted below.

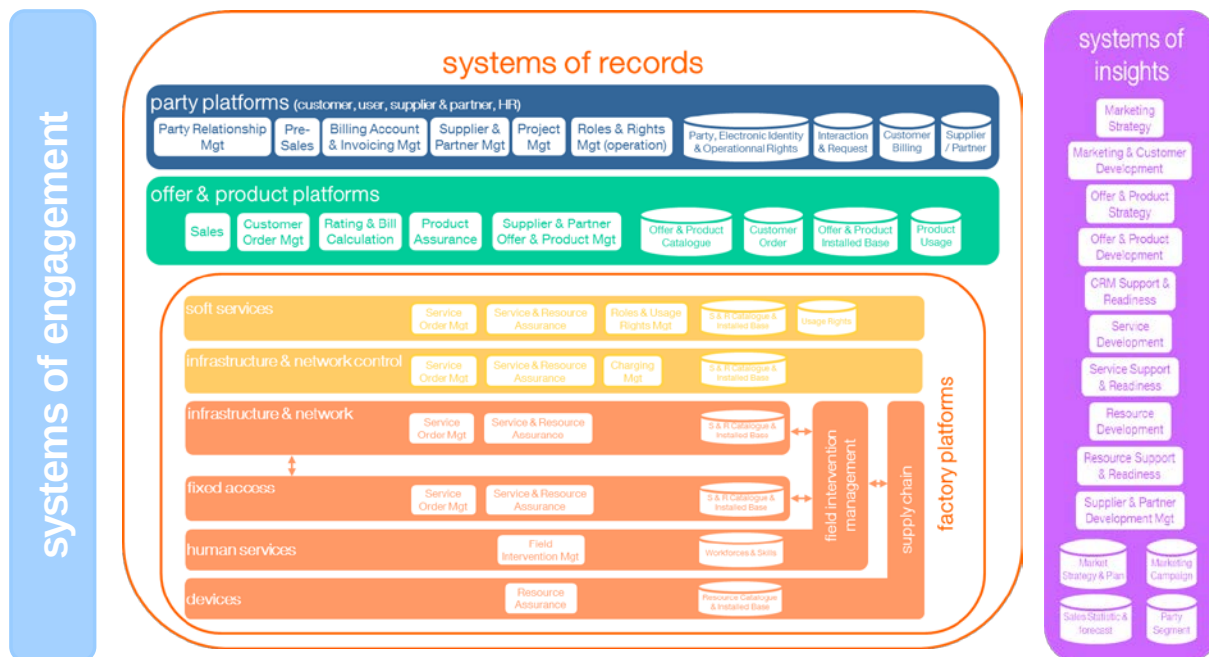


Figure 26 – Orange's business functionality

In Orange architecture:

- **Systems of Engagement** deals with User Interfaces (UI) sequences to carry out Party Interactions using APIs. They hold no business process. They are tied to fast evolving technologies to support Front End portals and Secured APIs exposed to 3rd party systems.
- **Systems of Record** deals with operational processes. They handle Offers & Products (WHAT?) built, run and maintained in a Factory (HOW?) that provides the underlying Services (aka CFS) or physical products, involving Parties (WHO & WHY?).
- **Systems of Insights** deal with all non-operational and Analytical processes and support the strategy design. They process information produced by the enterprise to provide various insights on its activity (including BI, Financial Management, Fraud Management & Revenue Assurance,...). They perform complex event/data processing (e.g. eligibility assessment) and interwork with other systems through event publication.

System of record is composed of 3 platforms:

- **Party Platform** handles Persons (employees, customers, users), Parties (suppliers, partners, organizations, external systems); it includes all related data and repositories and the associated management functions. This includes CRM, Invoicing, Financial Management and all Human Resources functions. This platform holds the functions related to the Identity. It is therefore involved (by a function call) every time a process requires personal identification
- **Offer & Product platform** handles the commercial catalogue of Offers and Products, the installed base (with the link to the contract owner in the Party platform). It manages all the function required for the Customer Order Management (order capture to bill calculation), Product assurance, Product rating, QoS management. This platform is kept agnostic of the customer, since the data and functions it carries do not depend on which party is involved.
- **Factory Platform** handles the technical plan on which the Products rely. It holds the asset of technical solutions (CFS and resources) facing the Products defined in the Offer & Product Platform catalogue. This platform is responsible for the delivery of the CFS requested by the Customer Order Management. It makes the choice of the appropriate solution (set of resources), delivers it and then operates, repairs the service. This platform holds the

technical repositories and installed Resource base (that is the installed CFS linked with the technical resources of the solution).

5.3.1. Level 0 of API Deployment View

The deployment view of the API's is aligned with the Party/Offer & Product/Factory view as the overall business abstractions as below:



Figure 27 – Orange's level 0 of API Deployment View

5.3.2. Level 1 of API Deployment View

In this Level 1 depiction, the API used by Orange are mapped to the platform layers above. They can be Orange Standard API (in black) or TM Forum standard API (in white).



Figure 28 – API used by Orange are mapped to the platform layers

In this Level 1 depiction, the API' from the API Map are mapped to the Orange platform layers.



Figure 29 – APIs are mapped to the platform layers

5.4. API Deployment View in Huawei

Huawei deployment view is composed of three main parts and infrastructure:

- Systems of Engagement are technologies that directly support customer interactions through software.
- Systems of Record are technologies that manage and store transactions, content, and processes.
- Systems of Insight are technologies that provide business analysis and statistics.
- Systems of Operation are technologies that intelligently manage physical products, networks, and infrastructure.

5.4.1. Level 0 of API Deployment View

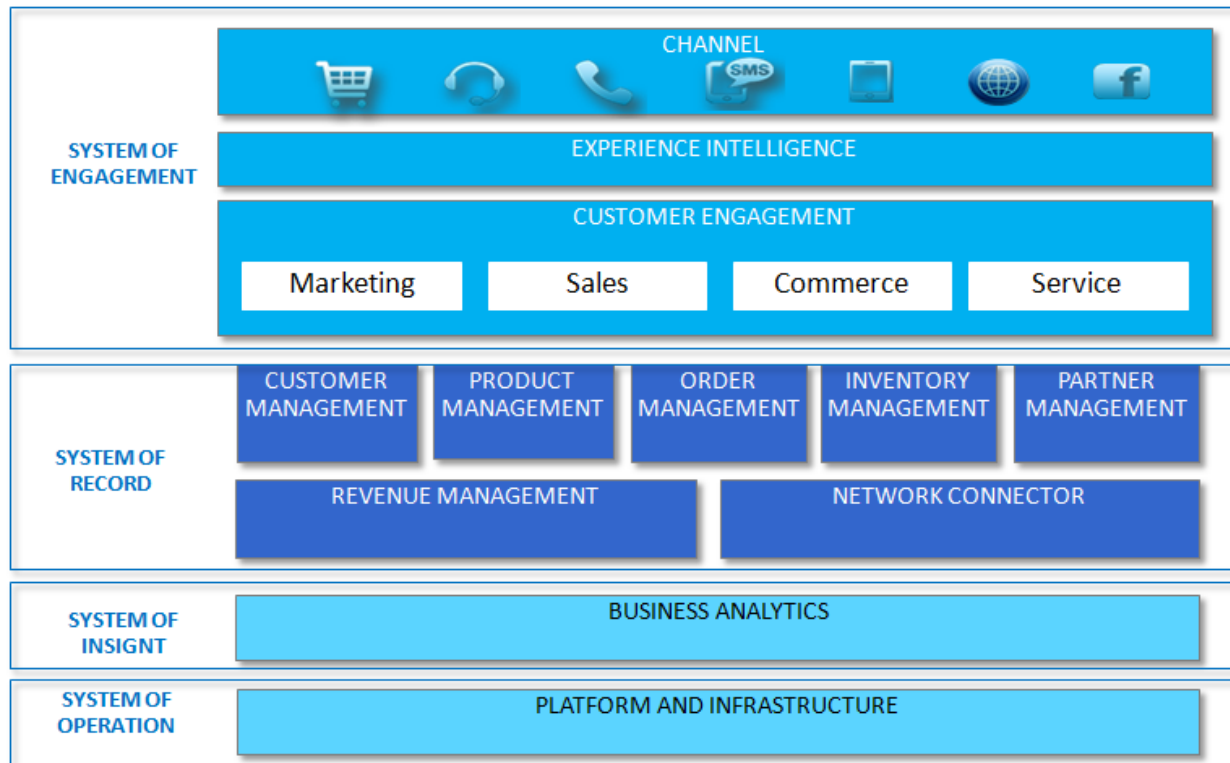


Figure 30 – Level 0 of Huawei's deployment view

5.4.2. Level 1 of API Deployment View

In this Level 1 depiction, the TMF APIs are mapped to the platform layers above.

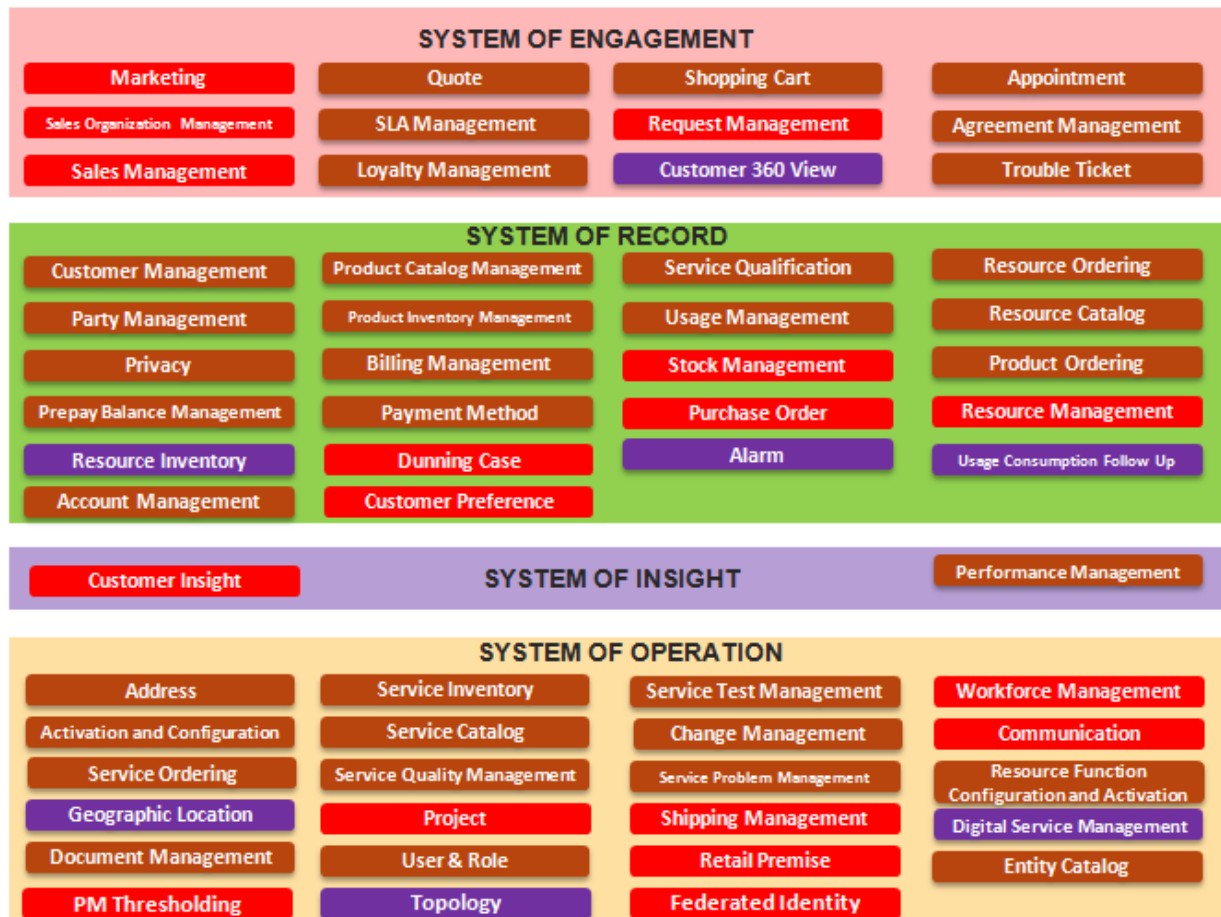


Figure 31 – Level 1 of Huawei's deployment view

6. API verification via Business Process (GB921D, PRELIMINARY)

The business process is the end to end line which organizes a group of activities for the certain purpose. When these activities collaborate together to support the accomplishment of the process, API is necessary for the interactions between the activities. By describing the API used in the steps of business process, the completeness and consistency of APIs can be verified.

There is wide range of business processes which are adopted in the different domains. The recommended selection for the process is based on TM Forum Process Framework (eTOM). In this document, API verifications will be built based on these published outcomes.

The Level 3 of TMF Process Framework is chosen as the benchmark of API verification. The Level 3 is the standard process which is also used in GB921D for process decomposition. The granularity of other levels appears improper because they are too high (Level 0/1/2) or too low (Level 4/5).

The section number of Level 2 and Level 3 process is kept in the mapping, so the original TM Forum Process Framework content can be conveniently found for comparison.

The TMF API relevance to the TMF Process Framework is depicted in three types:

✧ API name

If the Level 3 Process has the corresponding API to support, the name of API is shown in the mapping table.

✧ API is not defined

If the process may be supported using API from IT system, but the API has not been defined in the API Map, it is indicated as “API is not defined” so API can be considered in future.

✧ API is not required

If the process does not require API to support, for example, this process is manually handled, this process is indicated as “API is not required”

Note: this assessment is based on the knowledge until TMF R16.5. New API may be induced in future for the process which is currently recognized as “API is not required”

Please refer to the spreadsheet for the *GB992 Addendum Mapping between API and Business_Process_Framework*. This is the addendum of this document to contain all the details of the mapping to support the API verification.

7. Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

7.1. About this document

This is a TM Forum Guidebook. The guidebook format is used when:

- o The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.
- o Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.
- o Information about the marketplace is provided, as in the report on the size of the BSS/OSS market.

7.2. Document History

7.2.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	12-Jun-2016	Pierre Gauthier	First issue of document
1.0.1	13-Jun-2016	Alicja Kawecki	Updated cover; minor formatting/style edits prior to publication for Fx16
2.0.0	01-Dec-2016	Pierre Gauthier	Updated for Fx16.5
2.0.1	09-Dec-2016	Alicja Kawecki	Updated cover, minor formatting/style edits prior to publication for Fx16.5
3.0	12-Jul-2018	Hongxia Hao	Updated Section 7 and 8 according to Release R17.5 and R18.0.
3.1	12-Jul-2018	Cécile Ludwichowski	Mapping with SID update by the SID / Open API mapping taskforce.
3.2.0	24-Jul-2018	Add inputs from Takayuki Nakamura (NTT) and Dirk Rejaki (Digiglu)	Update text for Resource Pool AIP (pg 43) and Digital Services API (pg 29)
3.2.1	01-Aug-2018	Adrienne Walcott	Formatting/style edits prior to R18 publishing

Version Number	Date Modified	Modified by:	Description of changes
3.2.2	24-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

7.2.2. Release History

Release Number	Date Modified	Modified by:	Description of changes
1.0	12-Jun-2016	Pierre Gauthier	First release of document
2.0	01-Dec-2016	Pierre Gauthier	Second release for Fx16.5
18.0.0	24-Jul-2018	Derek Flexer	Updated for R18
18.0.1	24-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

7.3. Company Contact Details

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